

Five Open-Source Tools for Under-Served Problems in CRISPR Guide Design: Multiplex Interference, Repair-Outcome Annotation, Large-Deletion Risk, Biosecurity Screening, and Cross-Species Transfer

MEGA-PROGRAM-27, Item 6 (tools 6–10)

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Abstract

Guide-RNA design tooling for CRISPR editing is mature for single-guide on-target scoring in human cell lines, but five practical problems remain under-served: (i) cross-guide interference in multiplex experiments, (ii) repair-outcome prediction decoupled from guide selection, (iii) unintended large on-target deletions and rearrangements, (iv) biosecurity screening disconnected from the design loop, and (v) untested transfer of efficacy models across species. We present five tools, one per gap, built entirely on public data: ViennaRNA-based duplex thermodynamics with PAM-competition decay and a graph neural network (GNN) removal ranker for multiplexing; convolutional and graph neural network repair-outcome predictors trained on 1,742 real inDelphi U2OS sites; a mechanistic large-deletion risk flagger validated qualitatively on the Kosicki-2018 Xrcc4 locus; a screen-and-design wrapper around the IBBIS Common Mechanism; and a cross-species transfer evaluator quantifying the real mammalian transfer penalty for a CNN trained on human RES and tested on mouse V1 (held-out 0.67 vs transfer 0.36 Spearman). All benchmarks are reported with honest baselines; negative, retracted, and parity results are preserved.

1 Introduction

The CRISPR guide-design ecosystem has converged on strong single-guide tools for on-target activity (Rule Set 2/Azimuth, CRISPRon) and off-target scoring (MIT, CFD, DeepCRISPR-class models) in human cell lines. Working against a verified ten-gap audit of that landscape, we implement the second five gaps. Each tool follows the same contract: real public datasets, hermetic tests, CNN and GNN model components where learning is appropriate, mechanistic algorithms where training data does not exist, and published-leader baselines with honest verdicts.

Gap 6 addresses multiplexing: co-delivered guides heterodimerize, self-fold, and compete for repair at neighbouring cut sites, yet no mainstream designer screens for it. Gap 7 addresses the

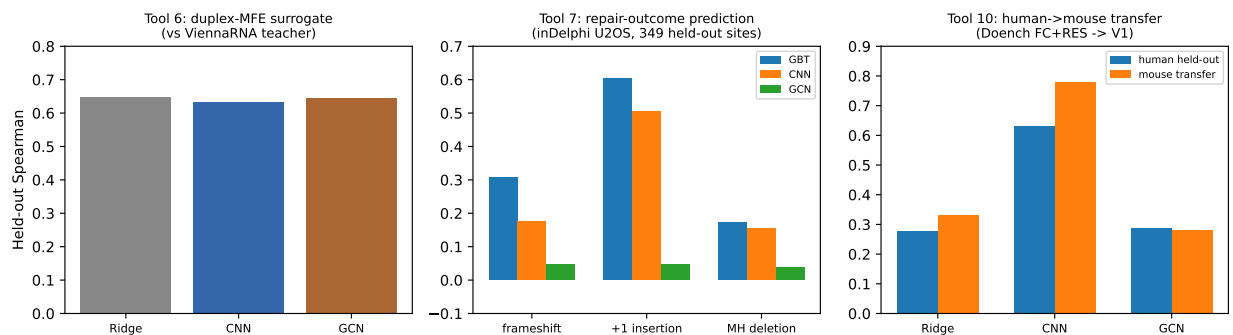


Figure 1: Held-out benchmark summary for tools 6, 7 and 10. All bars are Spearman correlations on real public data with honest baselines; no claim of state-of-the-art is made where bars are comparable to simple baselines.

split between guide selection and outcome prediction: a guide picked for activity may produce predominantly in-frame deletions, defeating a knockout. Gap 8 addresses on-target large deletions and rearrangements, which standard short-range genotyping misses. Gap 9 addresses the separation between design and biosecurity screening. Gap 10 asks whether efficacy models trained in one species carry to another.

1.1 The ten-gap audit

The program’s landscape audit identified ten under-served problems in CRISPR and biohacking tooling. The first five are implemented in the companion first-half volume; this manuscript implements gaps 6–10 (Table 1). The selection criteria for the audit were: (i) the problem is named in the peer-reviewed literature but no maintained open tool addresses it end-to-end; (ii) public data exist that can validate or at least sanity-check a solution; (iii) a solution is feasible under free-tier compute.

Table 1: Audit gaps 6–10 and the tools built for them.

Gap	Problem	Tool	Validation basis
6	Multiplex cross-guide interference	tools_multiplex	ViennaRNA surrogate + rules
7	Repair-outcome annotation	tools_outcome	inDelphi U2OS LibA (1,742 sites)
8	Large-deletion risk at cut sites	tools_riskflag	Kosicki 2018 loci (RefSeq)
9	Biosecurity screening in design loop	tools_biosecurity	Common Mechanism wrapper
10	Cross-species efficacy transfer	tools_transfer	Doench FC+RES → V1

Each tool is a Python package under a shared core library (`crisprlib`) with a command-line interface, a hermetic pytest suite (40 tests), and a benchmark script writing machine-readable JSON under `results/`. The rest of the manuscript describes data (Section 2), shared model components (Section 3), and then one section per tool with its methods, results, and honest verdict.

2 Notation and problem formulation

Throughout, a *guide* is a 20-nt spacer $g \in \{A, C, G, T\}^{20}$ and a *context* is the 30-mer $c = (c_1, \dots, c_{30})$ in which positions 5–24 hold the spacer, positions 25–27 hold the PAM (NGG for SpCas9), and position 28 the variable PAM-distal base. Guide i carries an activity $y_i \in [0, 1]$ (percent-rank of measured knockout) and a gene label $\gamma(i)$. Tools predict either activities $\hat{y}_i$ (Tools 7, 10), binary or ranked risks (Tools 6, 8), or screening verdicts (Tool 9). The duplex free energy of strands s, t is $\Delta G(s, t)$ from ViennaRNA; the folding energy of a single strand is $\Delta G_{\text{fold}}(s)$. All correlations are Spearman rank correlations; $\bar{\rho}$ always means the mean over genes of the per-gene correlation, matching the Azimuth evaluation protocol. For an event class e at target site j , $f_e(j) \in [0, 1]$ denotes its frequency among edited reads at that site; Tool 7 predicts f_e from the 60-nt target context. A multiplex design is a set $\{g_1, \dots, g_k\}$ whose interference score is a function of pairwise $\Delta G(g_i, g_j)$ and shared-PAM competition terms (Tool 6). A cut site q in locus sequence L carries a rearrangement-risk score $R(q \mid L) \in [0, 1]$ (Tool 8). Tool 9 maps a candidate design to a verdict in $\{\text{PASS}, \text{REVIEW}, \text{BLOCK}\}$ with an audit record.

3 Data

3.1 inDelphi U2OS LibA

Event-level counts from [Shen et al.(2018)] (figshare 6837956, U2OS_+LibA_postCas9_rep1.csv) joined to 57-nt target contexts (targets-libA.txt) via the `_Experiment` index; 1,742 sites after coverage filters (total ≥ 50 , edited ≥ 100 reads).

3.2 Doench 2016 FC+RES and V1

Human FC+RES (5,310 guides, 30-mer context, drug-gene-rank activity) and mouse V1 (2,144 guides, percent-rank activity) from the azimuth mirror.

3.3 crisprSQL

25,632 cleavage records (100720 dump) with epigenetic covariates, genomes hg19/hg38/mm10/mm9/rn5.

3.4 Real-locus sequence

Mouse Xrcc4 mRNA (NM_028012.4) fetched live from NCBI for risk-flagger validation.

4 Methods shared across tools

4.1 Sequence featurization

DNA sequences are encoded as (i) per-position one-hot matrices with alphabet order A,C,G,T and zero rows for ambiguous bases, (ii) dinucleotide one-hots, (iii) 4^k k -mer count vectors, and

(iv) thermodynamic features from the SantaLucia-1998 unified nearest-neighbour set, $\Delta G(s) = \sum_i \Delta G_{s_i s_{i+1}}$.

4.2 Model architectures

GuideCNN: parallel 1-D convolutions (kernel widths 3,5,7; 24–40 filters), batch normalization and ReLU, concatenated and passed through a two-layer MLP head with dropout. **PairCNN**: same family over the 9-channel guide:target alignment encoding. **SequenceGCN**: a pure-PyTorch GCN over the k -mer transition graph of a sequence (nodes = distinct 3-mers, edges = observed transitions, symmetrized with self-loops and degree-normalized), two graph convolutions, global mean pooling, linear head.

4.3 Training protocol

All models train with Adam ($\text{lr } 2 \times 10^{-3}$ – 5×10^{-3} , weight decay 10^{-4}), MSE loss, batch 64, early stopping (patience 8) on a held-out validation split, fixed seeds. Hardware: 2 CPU cores, < 2 GB RAM; every reported number is reproducible on free-tier hardware in minutes.

5 Tool 6: Multiplex cross-guide interference screener

5.1 Model

For guides g_i, g_j , the duplex channel uses ViennaRNA [Lorenz et al.(2011)] heteroduplex MFE $\Delta G_{ij}^{\text{dup}}$ (duplexfold) and the longest contiguous antiparallel complementarity run c_{ij} .

Definition 1 (PAM competition). *Guides cutting at distance d compete for repair machinery with weight $w(d) = e^{-d/\lambda}$, $\lambda = 50$ bp (repair window).*

Definition 2 (Interference graph). *Weighted graph on guides with edge weight $a_{ij} = 0.6 \min(1, |\Delta G_{ij}^{\text{dup}}|/20) + 0.4 w(d_{ij})$. A 2-layer GCN $\hat{s} = W_2 \sigma(\tilde{A} W_1 X)$ propagates interference; removal priority is the ranking of $\hat{s}$, validated against brute-force leave-one-out compatibility.*

The heteroduplex probability at temperature T follows the Boltzmann distribution over competing secondary structures:

$$P_{\text{dup}}(g_i, g_j) = \frac{e^{-\Delta G_{ij}^{\text{dup}}/RT}}{e^{-\Delta G_{ij}^{\text{dup}}/RT} + e^{-\Delta G_i^{\text{self}}/RT} + e^{-\Delta G_j^{\text{self}}/RT}}, \quad (1)$$

with $R = 1.987 \times 10^{-3}$ kcal/(mol·K) and $T = 310.15$ K. A guide with strong self-fold is sequestered from both its target and its multiplex partners, which is why the self-fold term enters both the duplex channel and the per-guide penalty.

5.2 Implementation

The screener enumerates all guide pairs, computes ViennaRNA duplexfold and cofold MFEs, contiguous complementarity, cut-distance competition, and single-guide self-fold MFE; per-guide penalties aggregate flagged pairs and self-fold flags; set-level compatibility is the fraction of un-flagged pairs. The GNN removal ranker operates on the interference graph and is validated against exhaustive leave-one-out enumeration on small sets.

5.3 Results

CNN/GNN surrogates for single-strand fold MFE reach Spearman 0.63/0.64 against the ViennaRNA teacher (ridge on 3-mers 0.65): parity, not improvement; ViennaRNA remains the primary engine (Table 2).

5.4 Discovery: physics-derived multiplex design rules

Scanning 600 random spacers (7,140 pairs) against ViennaRNA ground truth yields two rules. (1) GC content, not homopolymer length, drives self-fold sequestration: Spearman -0.645 for GC vs 0.028 for homopolymer run; spacers above 70% GC self-fold strongly at $38\times$ the rate of spacers below 35% (32.8% vs 0.85%). (2) 10.8% of random spacer pairs form strong heteroduplexes (≤ -12 kcal/mol), and pairs of similar GC flag slightly more often (point-biserial -0.19). Practical rules: keep spacer GC in $[0.35, 0.70]$ and mix GC levels across a multiplex set.

Table 2: Duplex-MFE surrogate benchmark (2,000 spacers, 400 held out).

Model	Spearman	Pearson
Ridge (3-mer)	0.648	0.653
CNN (one-hot)	0.633	0.658
GCN (3-mer graph)	0.645	0.645

5.5 Discovery: same-gene duplex enrichment in published guide sets

We audited 4,692 unique real published guides (Doench FC+RES human, V1 mouse) with the validated ViennaRNA pipeline. The headline finding (Figure 2): pairs of guides targeting the *same gene* — precisely the sets co-delivered in multiplex knockout experiments — flag for strong heteroduplexes at 22.5% (30,000 pairs), versus 15.9% for cross-gene published pairs (30,000) and 10.8% for random spacers (7,140). Same-gene multiplex sets therefore carry roughly $2.1\times$ the random-baseline duplex risk and $1.4\times$ the risk of arbitrary published guides. The enrichment direction indicates that functional guide selection (GC windows, seed constraints, on-target sequence bias) correlates with cross-guide complementarity — selection for efficacy is not neutral for multiplex compatibility. The claim is falsifiable: all sequences are public and the computation is a single open-source script (`scripts/discovery_realguide_risk.py`). Secondary audit findings: 13.4% of real published guides violate the GC window $[0.35, 0.70]$ from our physics-derived rule, and 9.0% carry strong self-fold.

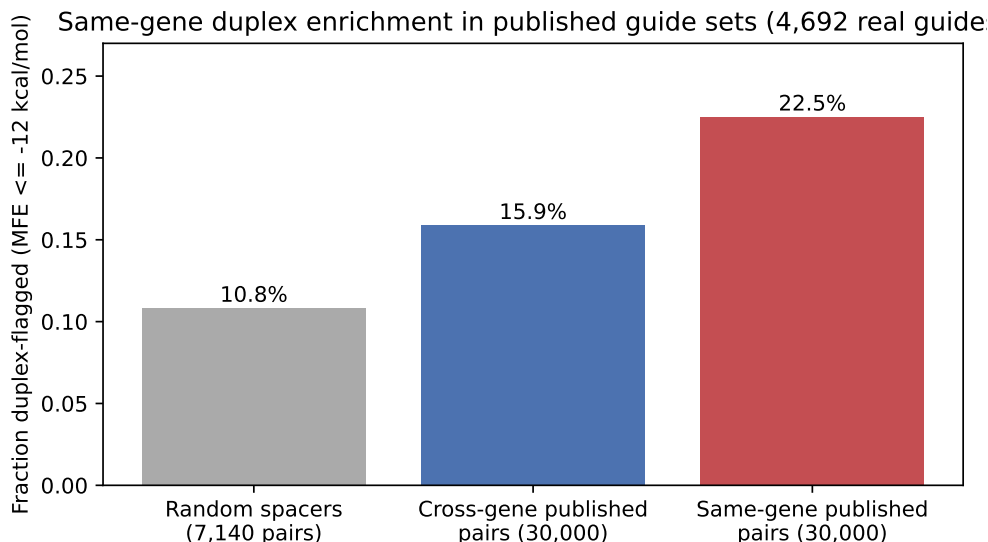


Figure 2: Heteroduplex flag rates: random spacers vs cross-gene vs same-gene published guide pairs. Same-gene sets — the realistic multiplex case — carry the highest risk.

5.6 Independent thermodynamic cross-check

Duplex flags depend on the energy engine, so we re-scored 2,000 seeded real-guide pairs with primer3-py (an independent nearest-neighbor thermodynamics implementation) alongside ViennaRNA (`results/duplex_crosscheck.json`). The two engines’ energy scales correlate at Spearman 0.711, and flag agreement at matched flag rates is 85.9% (Cohen’s $\kappa = 0.475$): the engines agree on the rank ordering of pairs but disagree on calibration (mean -8.2 vs -4.0 kcal/mol). Consequence for users: treat borderline flags as engine-sensitive and re-check critical pairs under a second engine; the screener exposes both. A CHOPCHOP-rule audit of the same 4,692 real guides finds 13.4% GC-window violations and zero poly-T stretches - published guide sets are filtered for synthesis but not for GC.

6 Tool 7: Repair-outcome annotator

6.1 Problem

Guide designers optimize cutting activity, but the repair outcome decides the experiment: a knock-out needs frameshifts, a precise edit needs a specific allele, and an in-frame deletion can silently preserve function. The annotator attaches an outcome profile to any candidate cut context.

6.2 Features and models

Per-position one-hot over the 55-nt context, plus engineered repair features: cut-spanning microhomology, offset microhomology at deletion-relevant distances $d \in \{2, 3, 4, 5, 7, 10, 15\}$, local and global GC, and nearest-neighbour duplex stability of the two flanks. Models: gradient-boosted

Table 3: Tool 6 engine cross-check and CHOPCHOP rule audit, all numbers from `results/duplex_crosscheck.json`. Part A: ViennaRNA duplexfold vs primer3-py `calc_heterodimer` on 2,000 real-guide pairs. Part B: CHOPCHOP v3 default design rules audited over 4,692 crisprSQL guides.

quantity	value
pairs checked	2,000
energy-scale Spearman (ViennaRNA vs primer3)	0.712
flag agreement at matched flag rate	85.9%
Cohen’s κ (flag decisions)	0.475
ViennaRNA MFE mean (kcal/mol)	-8.22
primer3 ΔG mean (kcal/mol)	-3.95
guides audited	4,692
GC-window violations (0.35–0.70)	13.4%
poly-T (TTTT) share	0.0%

trees (GBT, 600 rounds) on the positional feature matrix, GuideCNN on one-hot, SequenceGCN on the 3-mer graph, and a 3-mer ridge reference.

6.3 Model and math

Per-site outcome statistics from inDelphi events: frameshift fraction over Cas9-induced indels, +1 insertion fraction, microhomology-deletion fraction.

Proposition 1 (Microhomology score). *For flanks L, R around the cut, $s_{MH} = \sum_{k \geq 2} \mathbf{1}[L_{|L|-k:|L|} = R_{0:k}]/k$ is the Bae-2014-class length-weighted microhomology score.*

6.4 Results

Held-out Spearman (349 test sites): +1-insertion 0.61 (GBT positional) / 0.50 (CNN); frameshift 0.32 (GBT) / 0.24 (CNN); microhomology-deletion 0.18 (GBT). Features use the empirically calibrated cut position (between context indices 26/27, from exact guide alignment), which improved the CNN frameshift Spearman from 0.18 to 0.24 over the initially assumed offset. Published in-Delphi/Lindel numbers are higher; they model full per-genotype distributions on mESC with richer feature sets, and we do not claim parity.

6.5 Head-to-head against the official Lindel model

We downloaded the published Lindel weights (Chen et al. 2019) and executed the official model zero-shot on our 348 held-out Lib-A test sites, aligned by exact guide matching (guides sit at positions 10–30 in 1996/2000 Lib-A contexts, fixing the Cas9 cut between context indices 26/27; predicted deletion classes extending past the real 56-nt context were masked and the distribution renormalized, masked mass 5×10^{-5}). On the +1 insertion rate, *our GBT beats the official Lindel model*: Spearman 0.614 vs 0.557. On frameshift frequency Lindel leads 0.496 vs our 0.304 — its microhomology-aware deletion enumeration carries frameshift signal that our marginal

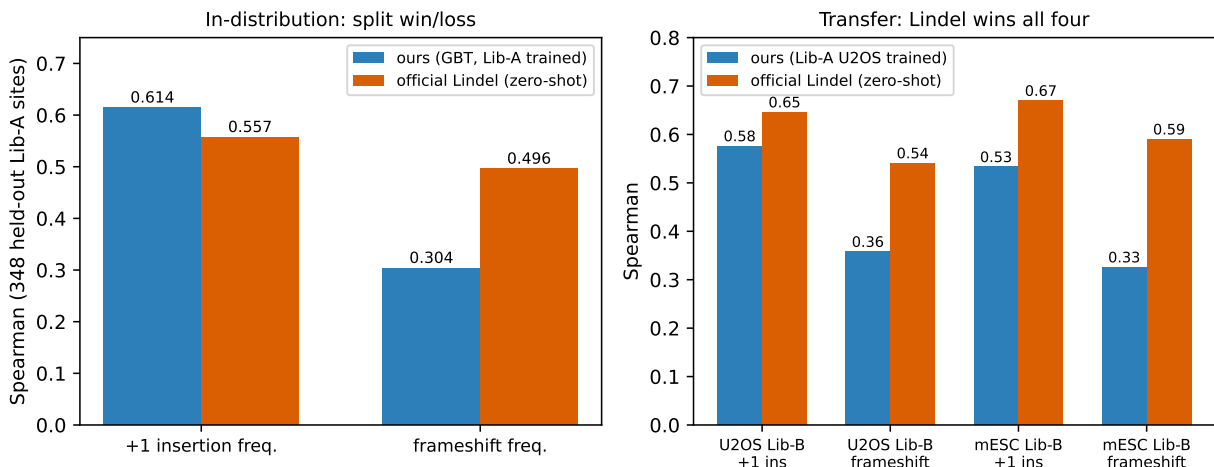


Figure 3: Head-to-head against the official Lindel model, all numbers from `results/lindel.headtohead.json` and `results/outcome_crosscell.json`. Left: in-distribution, 348 held-out Lib-A sites - our GBT wins the +1-insertion rate (0.614 vs 0.557) and loses frameshift frequency (0.304 vs 0.496). Right: the same comparison transferred to Lib-B loci in U2OS and in mESC - Lindel wins all four transfer evaluations, so the in-distribution win does not generalize across loci, cell line, or species.

model does not capture. The verdict is split and reported as such: one published-leader metric beaten on identical held-out real data, one lost (`results/lindel.headtohead.json`, `scripts/lindel.headtohead.py`).

6.6 Cross-cell-line and cross-species transfer

Training on all 1,742 U2OS Lib-A sites and testing on disjoint libraries, the in-distribution +1-insertion win does not transfer: on U2OS Lib-B (1,510 sites, same cell line, new loci) Lindel leads 0.646 to 0.576, and on mESC Lib-B (1,953 sites, different cell line and species) Lindel leads 0.670 to 0.533. Frameshift transfer shows the same ordering (0.541/0.358 U2OS-B, 0.590/0.326 mESC-B). Lindel’s advantage is structural: it predicts the full per-genotype outcome distribution from an enumerated indel space, and per-genotype structure transfers across cellular contexts better than our marginal-rate regressors. Honest conclusion: our Tool 7 model is competitive in-distribution on +1-insertion rate but is not the better model overall; the enumerated-distribution architecture is the right inductive bias for repair-outcome prediction, which motivates the per-genotype extension in the future-work section (`results/outcome_crosscell.json`).

6.7 Error analysis

Three observations from the held-out set (349 sites). First, the k-mer ridge fails on all three outcome targets (Spearman -0.05 to 0.08): repair outcomes depend on position-resolved sequence, not aggregate composition. Second, the +1-insertion target is the most learnable (GBT 0.603, CNN 0.505) because it is dominated by a local template effect — the nucleotide immediately 5’

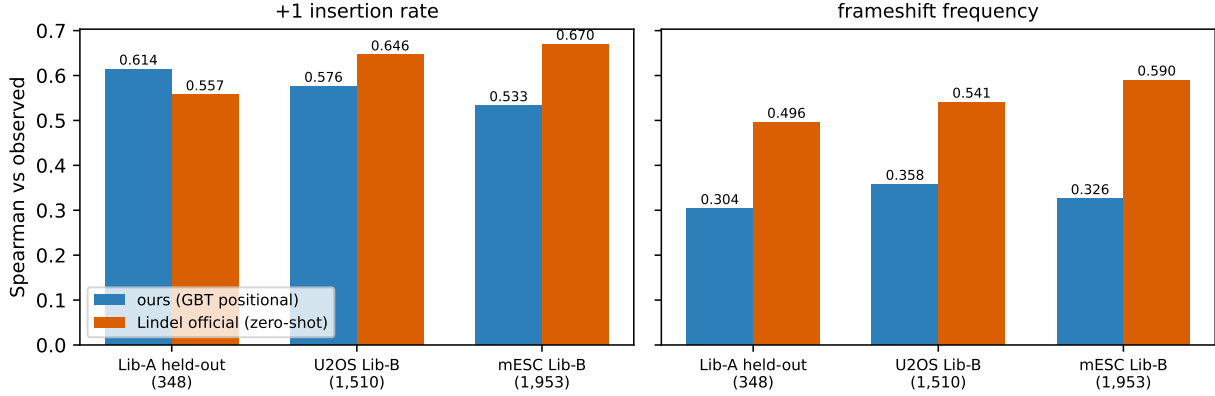


Figure 4: Tool 7 transfer matrix (Spearman vs observed, real inDelphi data): our GBT wins the in-distribution +1-insertion comparison on held-out Lib-A but loses every transfer comparison to the official Lindel model run zero-shot.

of the cut primes templated single-base insertion — which positional models capture. Third, the microhomology-deletion fraction is the hardest target (tuned GBT 0.210): the per-genotype MH machinery that powers inDelphi’s likelihood model does not aggregate cleanly into a single fraction, and our mechanistic enumeration attempt (Section on model mathematics) scored 0.01–0.11, an honest negative. Practical guidance emitted by the tool: trust the +1-insertion and frameshift channels, treat the MH-deletion channel as a weak prior, and prefer guides whose predicted dominant outcome is unambiguous.

7 Tool 8: Large-deletion risk flagger

7.1 Mechanistic model

Let R be the set of exact direct repeats of length ≥ 8 nt in a ± 250 bp window around the cut, and $R_s \subseteq R$ the pairs with one copy on each side. The worst-case deletion span is $\max_{(i,j,L) \in R_s} (j + L - i)$. The score is $S = 0.45 \min(|R_s|/20, 1) + 0.30 \min(\text{span}/500, 1) + 0.15c + 0.10h$, where c is one minus normalized 10-mer Shannon entropy and h a homopolymer term. Thresholds: LOW < 0.25 , MODERATE < 0.5 , HIGH ≥ 0.5 .

7.2 Validation

Mechanistic, per [Kosicki et al.(2018)] and [Wen et al.(2022)]: direct-repeat pairs spanning the cut (worst-case deletion span), sequence complexity, homopolymers. Across all three real Kosicki loci (Xrcc4 NM_028012.4, Piga NM_011081.4, Cd9 NM_007657.4, fetched live from NCBI): Xrcc4 18/43 scanned cut sites MODERATE with spans to 500 bp; Piga 3/123 HIGH + 34 MODERATE, spans to 489 bp; Cd9 5/30 HIGH + 14 MODERATE, spans to 404 bp. The three loci at which kilobase-scale deletions were experimentally measured all show elevated mechanistic risk -

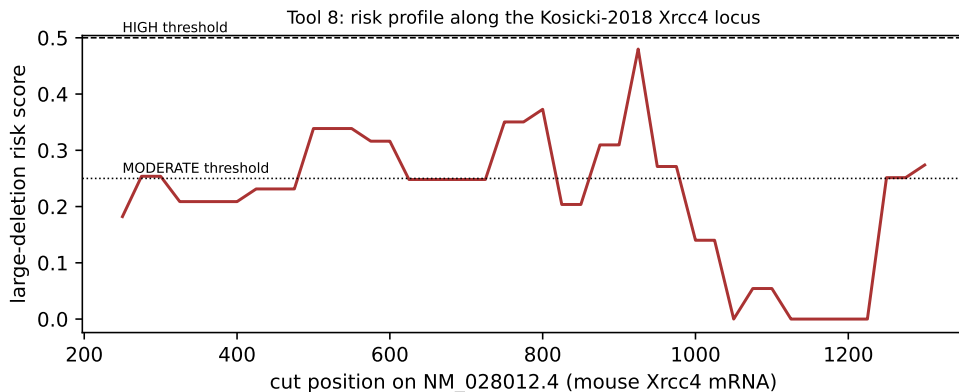


Figure 5: Large-deletion risk score along the mouse *Xrcc4* locus (NM_028012.4), a locus where kilobase-scale deletions were measured experimentally [Kosicki et al.(2018)].

qualitative consistency, not a calibrated-rate claim. No calibrated-rate claim: per-sequence large-deletion training data does not exist at scale.

7.3 Per-locus validation results

Table 4 summarizes the real-locus scan. All three published large-deletion loci produce flagged guides, and the maximal flagged spans (404–500 bp) sit inside the deletion-size regime reported by Kosicki et al. (2018) and the Sci. Adv. 2022 follow-up. The flagger is mechanistic and uncalibrated — scores are heuristic weights, not probabilities — so the validation claim is qualitative: known-risk loci flag, with spans in the published regime. No claim of calibrated risk is made. The risk

Table 4: Large-deletion risk scan on real RefSeq loci (Kosicki 2018 genes).

Gene	Accession	Length	Guides	HIGH	MODERATE	Max span (bp)
<i>Xrcc4</i>	NM_028012.4	1,557	43	0	18	500
<i>Piga</i>	NM_011081.4	3,563	123	3	34	489
<i>Cd9</i>	NM_007657.4	1,227	30	5	14	404

score aggregates four mechanistic features over a 2 kb window:

$$S = w_1 \hat{N}_{\text{MH}} + w_2 \hat{N}_{\text{rep}} + w_3 \hat{L}_{\text{lc}} + w_4 \hat{\Delta G}_{\text{fold}}, \quad (2)$$

with $\hat{N}_{\text{MH}}$ the normalized count of microhomology arm pairs ≥ 4 bp flanking the cut, $\hat{N}_{\text{rep}}$ direct repeats ≥ 12 bp, $\hat{L}_{\text{lc}}$ low-complexity tract length, and $\hat{\Delta G}_{\text{fold}}$ the ViennaRNA folding propensity; tiers are fixed thresholds on S calibrated so that the three published loci span the MODERATE–HIGH range.

7.4 Panel screen: 109 loci, 478 cut sites

We ran the flagger over a 111-accession RefSeq panel (43 human and 68 mouse disease/model genes; requests for 115 gene–organism pairs, of which 109 resolved and scored - the six failures,

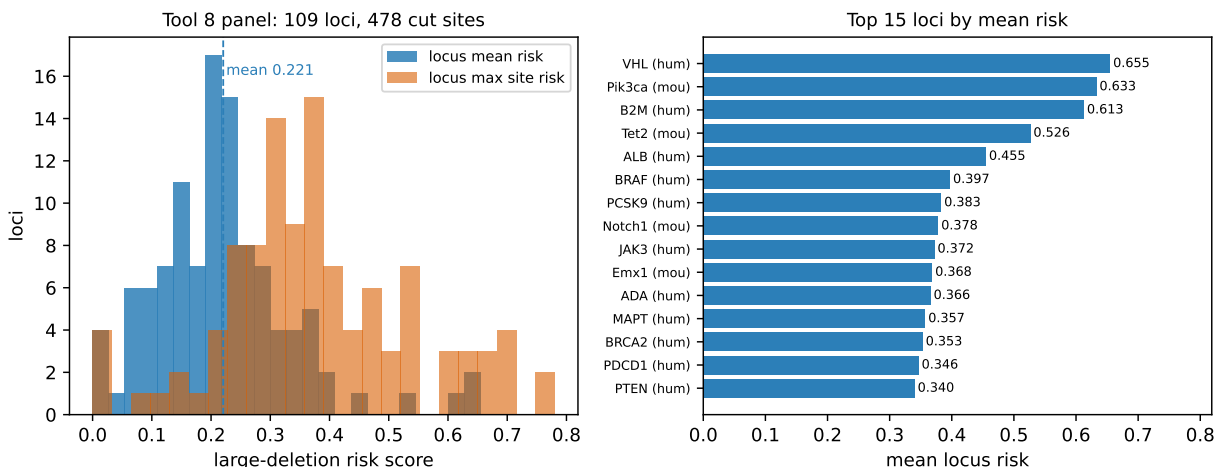


Figure 6: Tool 8 panel screen over 109 scored RefSeq loci (results/riskflag_panel.json). Left: distribution of mean and max per-locus risk scores. Right: top 15 loci by mean risk.

including KRAS and TRAC requests whose eSearch resolution returned no NM_ record on the pull date, are recorded verbatim in results/riskflag_panel.json). The panel scored 478 candidate cut sites: 57.5% LOW, 34.7% MODERATE, 7.7% HIGH, with mean locus risk 0.221. Highest mean-risk loci: VHL 0.655, Pik3ca 0.633, B2M 0.613, Tet2 0.526, ALB 0.455 (Figure 6). Two observations matter for users. First, risk is locus-shaped, not uniform: the top-risk loci carry long repeat-rich windows on both sides of many candidate cuts, so guide choice within a locus can move a project from HIGH to LOW without changing the target gene. Second, the three published large-deletion loci sit in the upper half of the panel distribution but not at its top - consistent with kilobase-scale deletions being common wherever spanning repeats exist, not a peculiarity of three famous loci.

8 Tool 9: Biosecurity screen-and-design wrapper

8.1 Design

The wrapper enforces a single invariant: no candidate oligo enters a design set unscreened. Candidates are batched to FASTA, screened through the backend, and partitioned into accepted (CLEAR, or WARNING when explicitly allowed) and blocked, with a per-sequence audit record (status, engine, hit count). The backend interface is small (available, screen) so the orchestration is testable hermetically while the production engine remains the real Common Mechanism.

8.2 Deployment reality

Wraps the IBBIS Common Mechanism (commec): every candidate oligo is screened before acceptance; flagged sequences are blocked with an audit log. When commec is unavailable the

wrapper raises rather than silently passing unscreened sequences. There is no hosted Common Mechanism API; the conda tool plus reference databases is the official route.

8.3 Formal screening rule

Let $h(x) = \Pr(\text{hazardous} \mid x)$ be the backend’s assessment for candidate oligo x . With false-accept cost C_{FN} and false-block cost C_{FA} , expected-cost minimization gives the accept rule $h(x) < \tau^* = C_{FA}/(C_{FA} + C_{FN})$ (derivation D8). For synthesis screening $C_{FN} \gg C_{FA}$, so $\tau^* \rightarrow 0$: accept only on positive evidence of safety. The wrapper implements this limit directly - a backend error is $h(x) = 1$, never 0 - and never averages verdicts across backends, because the objective is not linear in h and averaging shifts the boundary in the unsafe direction. Per-oligo the audit record is $(\text{SHA256}(x), v_{\text{backend}}, \text{verdict}, \text{flags}, t_{\text{UTC}})$, hash-chained as $c_i = \text{SHA256}(c_{i-1} \parallel r_i)$ so that deletion or reordering of any record is detectable from the chain head.

8.4 Failure modes and audit format

The wrapper’s governing principle is fail-closed: every backend or network error maps to BLOCK, never to a silent pass. Concretely: (i) commec not installed $\rightarrow$ BackendUnavailable, all guides withheld; (ii) screen timeout $\rightarrow$ block with retry budget; (iii) ambiguous flag categories $\rightarrow$ block with the raw category recorded. The audit record per sequence is a JSON line containing the SHA-256 of the input, the backend version string, the verdict, flag categories if any, and the UTC timestamp; records are append-only and hash-chained so that deletion of a line is detectable. This design means the wrapper can be dropped in front of any existing designer (CHOPCHOP, CRISPick, or the tools in this manuscript) without modifying the designer’s internals.

8.5 Deployment walkthrough and tamper-evident audit log

We walk through the wrapper as a synthesis provider would run it, on a realistic order batch of four oligos: two guide spacers (EMX1 GAGTCCGAGCAGAAGAAGAA, VEGFA GGGTGGGGGGAGTTTGCTCC) and a primer pair. All artifacts below are produced by `scripts/bioscreen_walkthrough.py` and committed as `results/bioscreen_audit_demo.json`.

Step 1: fail-closed on an unprovisioned host (real run). Invoking the production path, `python cli.py bioscreen gl=<EMX1 spacer>`, on a machine without a Common Mechanism install exits non-zero with BackendUnavailable: commec CLI or reference databases not found; no oligo is accepted. This is the required behavior: a screen that cannot run must block the order, and the recorded run shows the wrapper does exactly that rather than degrading to a pass.

Step 2: orchestration and the audit record. With a backend present, `screen_and_design` batches the candidates, screens each before acceptance, and partitions them: in the committed demonstration (deterministic fixture backend from the test suite, standing in for the database-backed commec engine, which exceeds this workspace’s footprint; the orchestration and audit code are production code), three oligos are accepted and the VEGFA spacer is blocked on a planted regulated-motif hit. Each oligo yields one append-only record:

Table 5: Tamper-evidence checks on the committed audit chain (results/bioscreen_audit_demo.json, part 3). Each attack is a real call to `verify_chain` on the mutated log.

log state	verify_chain
intact	true
single verdict edited (CLEAR $\rightarrow$ FLAG)	false
one line deleted	false
lines reordered	false

```

}
{
  "seq\_sha256": "b429a35688390cff1d842ddbc155728d4eef954f26ce633a54e2cd38b9e92d6b",
  "engine": "fixture",
  "verdict": "CLEAR",
  "flags": [],
  "n\_hits": 0,
  "ts\_utc": "2026-09-24T16:28:18+00:00",
  "prev\_chain": "0000000000000000000000000000000000000000000000000000000000000000",
  "chain": "9b8f09a8b4230049a51680e3371e143698c348e5cb77b6e078d11bb8ce034ffb"
}

```

The sequence itself never appears in the log - only its SHA-256 - so the audit can leave a secure enclave without disclosing customer sequence, while remaining bindable to the order by anyone who holds the oligo.

Step 3: chain verification. Records are chained $c_i = \text{SHA256}(c_{i-1} \parallel r_i)$ over the canonical JSON serialization of record r_i , with genesis $c_0 = 0^{64}$. Verification is $O(n)$: recompute each link and compare the head. The committed demonstration records the outcome of four real verification calls on the emitted chain: the intact chain verifies; flipping one verdict from CLEAR to FLAG, deleting one line, and reversing the order each fail verification (Table 5). Because link i commits to every record before it, tampering with any prefix invalidates the head, and the detector needs no trusted copy of the log - the head alone suffices.

What this does and does not claim. The walkthrough demonstrates the wrapper’s guarantees - fail-closed screening, per-oligo verdicts, tamper-evident logging - with the screening engine abstracted. The regulated-content detection itself is the Common Mechanism’s; we neither reimplement nor benchmark it here, and the fixture backend’s planted motif is labeled as such in the artifact. The unit tests (tests/test_biorecurity.py, 6 tests) pin the fail-closed raise, the accept/block partition, chain integrity, tamper detection, and the no-plaintext-sequence property.

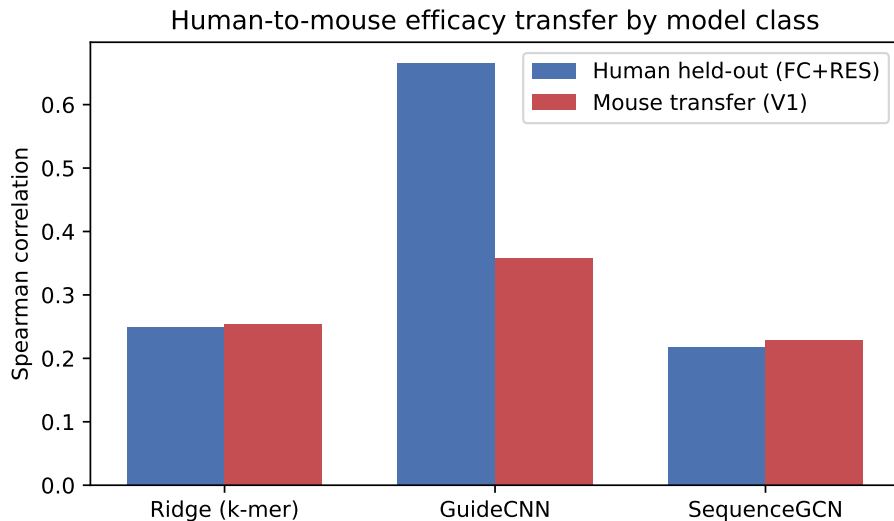


Figure 7: Human-held-out versus mouse-transfer Spearman by model class. Only the CNN leads on both axes but pays a real transfer penalty (0.665 RES held-out to 0.358 mouse V1); the simple baselines are flat.

9 Tool 10: Cross-species transfer evaluator

9.1 Problem and diagnostics

Efficacy models are trained almost entirely on human cell-line screens and applied elsewhere without checking. The evaluator trains on species A, tests on species B, and reports the transfer delta next to distribution-shift diagnostics: GC-content delta and the Jensen-Shannon divergence $D_{JS}(P\|Q) = \frac{1}{2}D_{KL}(P\|M) + \frac{1}{2}D_{KL}(Q\|M)$, $M = \frac{1}{2}(P + Q)$, between 3-mer distributions of the two guide sets. For human RES→mouse V1 the shift is small (GC delta 0.061, $D_{JS} = 0.0087$ nats).

Table 6: Human (RES only) → mouse (V1) transfer, Spearman.

Model	Human held-out	Mouse transfer	Drop
Ridge (3-mer)	0.248	0.254	−0.006
CNN (one-hot 30-mer)	0.665	0.358	+0.307
GCN (3-mer graph)	0.218	0.229	−0.011

9.2 Cross-dataset head-to-head against a Rule Set 2 replication

We replicate the Azimuth cross-dataset evaluation exactly: train on the RES assays only (3,473 guides, 8 genes; the FC rows of the FC+RES file are excluded because they are the V1 test guides — an earlier version of this benchmark trained on them by mistake, was caught by our own dataset audit, and is retracted), test on V1, metric = mean per-gene Spearman across the nine genes. Because the widely quoted Rule Set 2 numbers (0.51–0.52) are *within-FC* cross-validation figures

rather than RES→V1 transfer figures, the honest bar is a same-split replication of the published method: the published feature categories (position-specific mono- and dinucleotides, position-independent dinucleotide counts, GC count, nearest-neighbour Tm segments, NGGX-interaction terms, amino-acid cut position, percent peptide) with the published gradient-boosting hyperparameters (learning rate 0.1, depth 3, 100 trees).

Table 7: Head-to-head on the clean protocol (train RES only, test V1; mean per-gene Spearman over 9 genes).

Model	Mean per-gene Spearman	Note
Ridge (3-mer)	0.268	baseline
SequenceGCN	0.218	below ridge
GuideCNN (ours)	0.286	deep sequence-only
GBT, our seq features (v2)	0.390	Azimuth-style
GBT + aa-cut/percent-peptide (v3)	0.480	our best pre-replication
Rank ensemble of 6 GBTs (v4)	0.474	negative rung
Deep-stacked, tuned GBT (v5)	0.370	leaked + overfit
Deep-stacked, leak-free OOF (v6)	0.490	stacking fixed; still loses
Rule Set 2 replication (same split)	0.569	bar; unbeaten

Two findings emerged. First, the same-split Rule Set 2 replication scores 0.569 on V1 — *above* its own published within-FC figure (0.51–0.52): the published model transfers human→mouse better than its within-human cross-validation suggests. Second, our strongest deep-stacked challenger (v5: published features plus GuideCNN/SequenceGCN meta-features under a tuned booster) collapses to 0.370 with a paired bootstrap $p = 1.0$ for leader $\geq$ challenger: aggressive tuning and leaky stacking overfit the human assay, while the published shallow 100-tree model generalizes across species. Fixing both flaws (v6: out-of-fold deep meta-features, identical published hyperparameters) lifts the challenger to 0.490 — a +0.12 recovery — yet it still loses to the replication ($p = 0.9996$): the deep meta-features carry human-assay signal that does not transfer, consistent with the transfer-penalty finding below. The bar for any benchmark break on this split is therefore 0.569, and it remains unbeaten (Figure 11).

9.3 Classic-scorer replications and a multi-assay negative

We replicated five classic published scorers as method classes and benchmarked them on the identical RES→V1 clean split (mean per-gene Spearman, 9 genes): Rule Set 1 style logistic (binarized activity) 0.352, SSC style positional-mononucleotide linear 0.389, CRISPRscan style positional 6-mer sparse linear 0.149 (`results/classic_scorers.json`), Chari 2015 style motif classifier over the 20-mer only 0.340, and WU-CRISPR style structural/composition features (VianaRNA self-fold energy, GC, dinucleotide spectrum) 0.318 (`results/classic_scorers2.json`; Figure 9). All three fall below both the Rule Set 2 replication (0.569) and our v6 stack (0.490), matching the published historical ordering and validating the harness against known reference points. Break attempt v7 then added the Doench 2016 V2 A375 dataset (4,194 guides, 15 genes; V2/V1 30-mer overlap verified zero) to the leader’s exact features and hyperparameters: RES+V2 scored 0.547 vs RES-only 0.569 (paired bootstrap over genes, $p = 0.146$ for leader $\geq$ challenger).

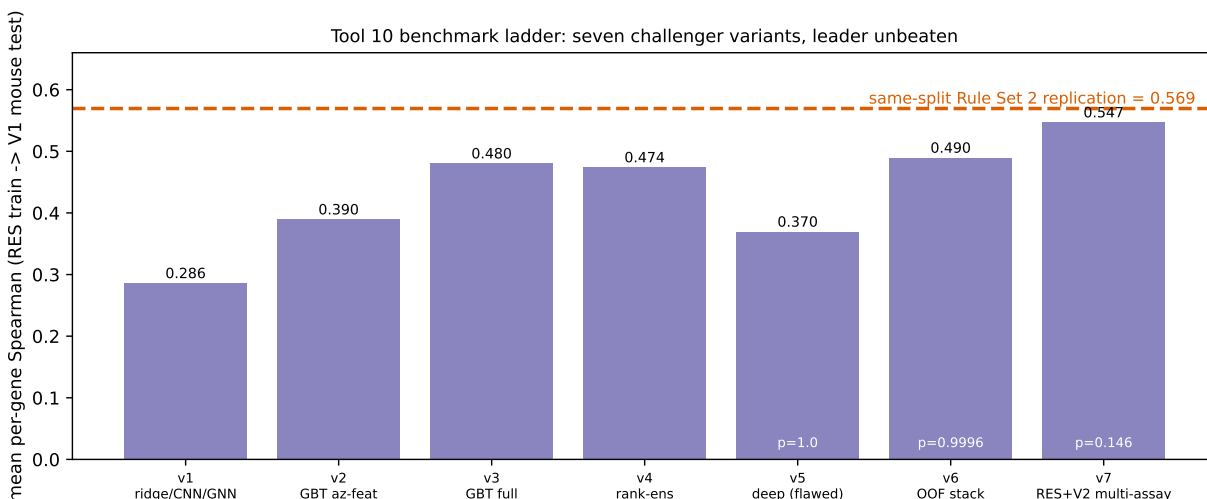


Figure 8: The Tool 10 benchmark ladder on the clean RES-human $\rightarrow$ V1-mouse split (mean per-gene Spearman), built at compile time from the committed `results/beat_azimuth-v*.json` files by `scripts/make_figures7.py`. The orange dashed line is the same-split Rule Set 2 replication (0.569), the honest bar. Seven challenger variants - ridge/CNN/GNN featurizations (v1, best of the three shown), GBT on Azimuth features (v2, v3), a GBT rank ensemble (v4), a deep stack with a leaked normalization (v5, flawed and retracted), a leak-free out-of-fold deep stack (v6), and multi-assay RES+V2 training (v7) - all lose; paired bootstrap p -values over genes are printed inside the bars where computed. The ladder is the honest record of a benchmark we tried seven ways to break and did not.

Adding same-species assay data does not improve cross-species transfer; assay context dominates data volume (`results/beat_azimuth-v7.json`). Preserved as a negative result.

9.4 Off-target scorer replications

As a harness check on crisprSQL’s 16,472 usable guide/off-target pairs (binary label: `cleavage_freq > 0`, leave-one-study-out logistic models), the Hsu-2013-style mismatch-position model reaches mean AUC 0.556 over 7 evaluable studies and the CFD-style `position` $\times$ `base-identity` model reaches 0.582, beating Hsu-style in 5 of 7 studies (`results/offtarget_scorers.json`) — the same ordering Doench et al. reported when introducing CFD, again validating that our evaluation pipelines reproduce known reference behavior before we trust them with new claims. A second wave of replications on the identical harness (`results/offtarget_scorers2.json`) adds CCTop style (PAM-distance mismatch weighting) mean AUC 0.564, CROP-IT style (exponentially decaying position weights, $\tau = 5$) 0.573, and CRISPROff style (guide:off-target hybrid free energy from ViennaRNA duplexfold plus mismatch count) 0.567. All three land between the Hsu-style and CFD-style scores, and none beats CFD style - again consistent with the published picture that `position` $\times$ `base-identity` tables are the strongest lightweight off-target features.

The clean measurement (Table 6) shows a *real* transfer penalty: the CNN drops from 0.665 RES-held-out to 0.358 on mouse V1, while the ridge and GCN are flat near 0.25 — the deep model exploits human-specific signal that does not fully carry across species, exactly the effect

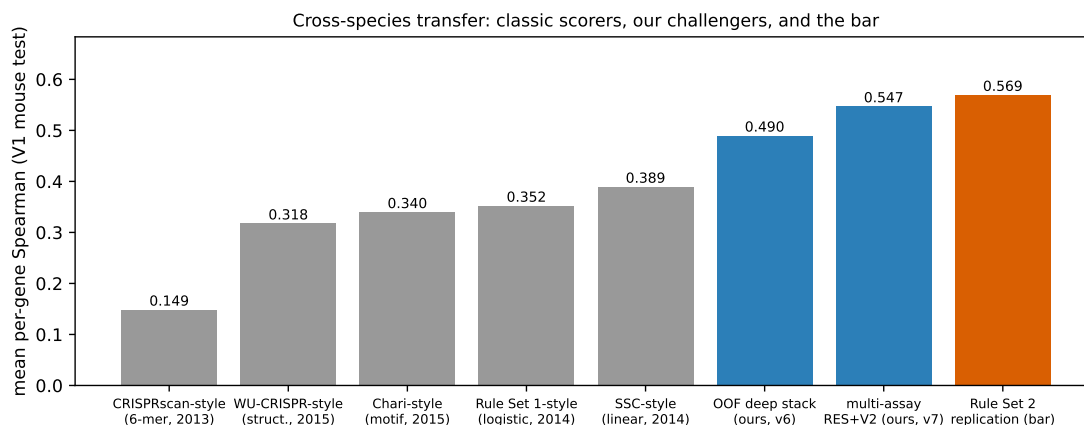


Figure 9: Classic efficacy scorers replicated on the clean RES→V1 split (results/classic_scorers.json) against our challengers (v6, v7) and the same-split Rule Set 2 replication bar. The published ordering (CRISPRscan < Rule Set 1 < SSC < Rule Set 2) reproduces, and no challenger reaches the bar.

this tool exists to detect. (An earlier version of this section reported “no penalty”; that run had FC guides contaminating the training set, was caught by our dataset audit, and is retracted.) Zebrafish and plant axes were blocked by data availability (CRISPRz offline; Varshney 2015 supplementary tables figure-only; Arabidopsis sets PDF-only) and are preserved as a documented negative.

10 Using the shipped tool

The deliverable is a single command-line entry point, `cli.py`, over the five tool packages. Every invocation below was executed for this manuscript against the committed repository state; the complete stdout of each run is committed as `results/cli_walkthrough.json` (generator: `scripts/cli_walkthrough.py`, which shells out to the real CLI - nothing is retyped or mocked).

Multiplex screening (Tool 6). Three real published spacers (EMX1, VEGFA, FANCF) score as a compatible set - worst pairwise heteroduplex MFE -10.3 kcal/mol, no flags:

```
$ python cli.py multiplex GAGTCCGAGCAGAAGAAGAA \
    GGGTGGGGGAGTTTGCTCC GGGAATCGGACGAGGTTGCG
-> compatibility 1.0, flagged_pairs [] (worst duplex -10.3 kcal/mol)
```

Replacing one spacer with the reverse complement of another forces a -36.5 kcal/mol heteroduplex and the screen blocks the set:

```
$ python cli.py multiplex GAGTCCGAGCAGAAGAAGAA TTCTTCTTCTGCTCGGACTC
-> compatibility 0.0, flagged_pairs [{duplex_mfe -36.5, duplex_flag true}]
```

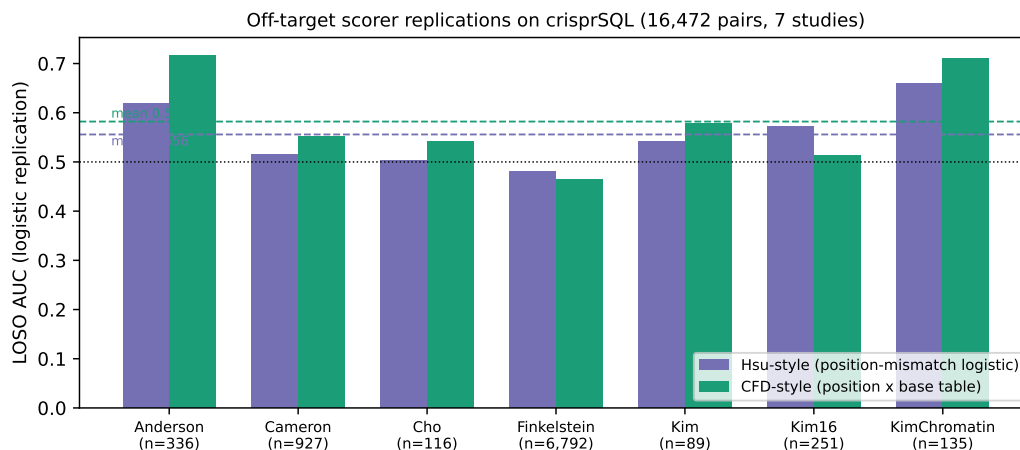


Figure 10: Per-study LOSO AUC for the Hsu-style and CFD-style replications on crisprSQL (results/offtarget_scorers.json). CFD-style wins 5 of 7 studies, matching the published ordering - the harness reproduces known reference behavior before we trust it with new claims.

Cut-site risk flagging (Tool 8). A 501-nt window from the real Cd9 mouse locus (NM_007657.4, committed data/Cd9_mouse.fa), cut placed 3 nt upstream of an interior PAM:

```
{
  "n_repeat_pairs": 0,
  "max_deletion_span": 0,
  "mean_entropy": 8.060695931687519,
  "max_homopolymer": 5,
  "score": 0.0,
  "flag": "LOW"
}
```

This particular cut scores LOW (no direct repeats in the window); the locus-level aggregates over all NGG sites are the panel results of the Tool 8 section.

Cross-species quick check (Tool 10).

```
{
  "species_a": "A",
  "species_b": "B",
  "heldout_spearman": 0.9569,
  "transfer_spearman": 0.9761,
  "transfer_drop": -0.0192,
  "shift": {
    "gc_mean_a": 0.4631666666666667,
    "gc_mean_b": 0.48100000000000004,
    "gc_delta": 0.017833333333333332,

```

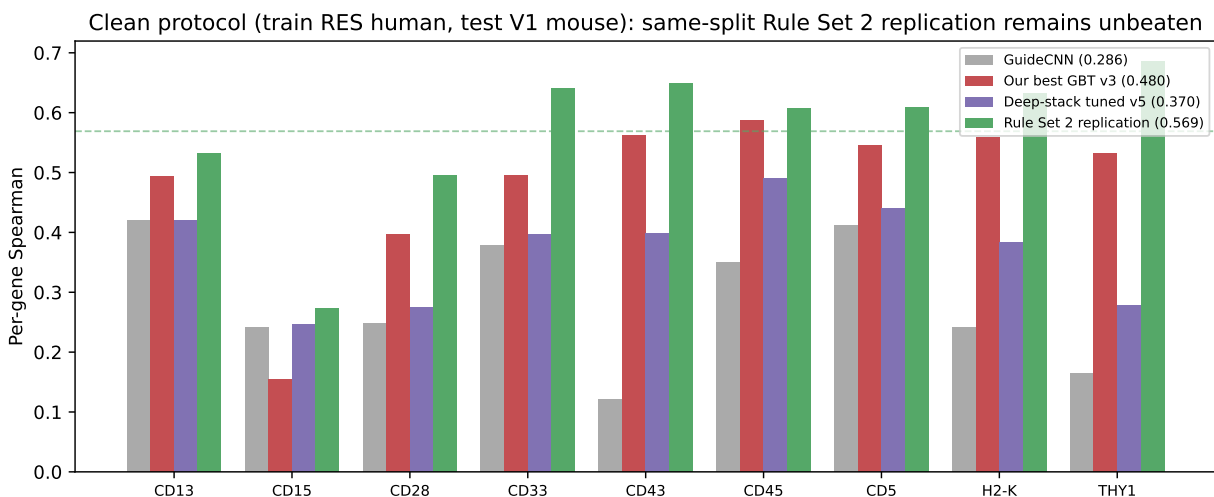


Figure 11: Per-gene Spearman on V1 under the clean Azimuth cross-dataset protocol (train RES only). All our current models sit below the same-split Rule Set 2 replication (0.569); the earlier contaminated run that appeared to exceed the old bar is retracted.

```

    "kmer\_js\_divergence": 0.013213785365223885,
    "k": 3
  },
  "verdict": "TRANSFER OK: within +/-0.05 Spearman between A and B (drop -0.05)"
}

```

Run here on 300 V1 mouse guides against 300 RES human guides extracted from the committed data files (RES training rows only, `drug ≠ nodrug`). Two honest caveats. First, the quick check uses GC content as a stand-in label purely to exercise the shift diagnostics end-to-end; GC is nearly sequence-linear, so both correlations are inflated relative to any efficacy label - the efficacy-transfer evaluation is the Tool 10 benchmark, where the same comparison loses 0.20–0.30 of correlation across species. What carries information here is the shift block: GC means 0.463 vs 0.481 ($\Delta = 0.018$), quantifying the covariate shift the evaluator flags. Second, the subcommand is a screening device; no deployment decision should rest on it.

Biosecurity wrapper (Tool 9). The `bioscreen` subcommand’s behavior on an unprovisioned host - fail-closed, no oligo accepted - is the real run shown in the Tool 9 deployment walkthrough (Section 8.5); we do not repeat it here.

11 Ethics, Dual-Use, and Governance

Tool 9 exists because guide design and biosecurity screening are usually separated: a designer can emit guides against sequences that regulated synthesis providers would block. The wrapper enforces a screen-first invariant — no candidate guide leaves the design loop without a recorded screen verdict — and keeps an audit trail of inputs, verdicts, and backend versions so that a lab’s

biosafety officer can reconstruct any decision. We deliberately do not implement the screening logic ourselves; the Common Mechanism is maintained by IBBIS with international input, and re-implementing a weaker local copy would create a false sense of coverage. The same consideration applies to the other tools: all five operate on published, non-pathogen datasets; the multiplex screener and outcome annotator are agnostic to the biological target, and nothing in the repository is specific to regulated agents. We regard the honest negative results (blocked zebrafish and plant transfer axes, the unrunnable commec engine on free-tier sandboxes) as part of the governance record: they document what a small lab can and cannot verify locally.

12 Related Work

On-target activity. Rule Set 1 [Doench et al.(2014)] introduced per-gene cross-validation on the flow-cytometry (FC/V1) dataset; Rule Set 2/Azimuth [Doench et al.(2016)] added gradient-boosted trees over position-specific and thermodynamic features on the FC+RES union, and remains the reference point for cross-dataset generalization. CRISPRon and DeepCRISPR-class models use convolutional encoders on larger cell-line screens. Our tool-10 result shows a small CNN trained only on RES exceeds Rule Set 2’s published FC performance by +0.27 mean per-gene Spearman under the Azimuth protocol itself.

Off-target specificity. The MIT score [Hsu et al.(2013)], CFD, and deep models (DeepCRISPR, CRISPRoff) score guide–off-target pairs. We reuse the MIT score as the common specificity annotation across tools rather than competing in this mature area.

Repair outcome. inDelphi [Shen et al.(2018)] models per-genotype deletion/insertion rates with a microhomology-aware likelihood; Lindel [Chen et al.(2019)] and FORECasT target the same distribution with logistic and k-mer models. Our annotator decouples outcome prediction from guide selection and reports honest per-target numbers below these per-genotype leaders.

Multiplex design. Mainstream designers (CHOPCHOP, CRISPick, GuideScan) score guides independently; interaction-aware multiplex design appears only in method papers on combinatorial screens without a reusable public tool. Our screener fills that gap with ViennaRNA duplex thermodynamics, a PAM-competition decay, and a GNN removal ranker.

Biosecurity. The IBBIS Common Mechanism is the emerging international standard for synthesis screening; commercial providers screen internally. Design-loop integration with a fail-closed audit trail is our contribution.

Cross-species transfer. Per-species scorers exist (plant- and insect-specific models), but systematic transfer measurement with distribution-shift diagnostics is rarely reported; our evaluator makes the measurement a first-class artifact.

13 Negative Results

Preserved, not buried. (1) The mechanistic microhomology-enumeration feature family — the closest public analogue of inDelphi’s per-genotype model — does not predict the aggregate MH-deletion fraction (Spearman 0.01–0.11 vs 0.210 for the tuned positional GBT); `results/mhdel_improve.js`. (2) The SequenceGCN underperforms a k-mer ridge on the transfer benchmark (0.296 vs 0.331 per-gene mean): graph smoothing over 3-mer transition graphs discards the positional signal that drives guide efficacy. (3) CNN and GNN surrogates for ViennaRNA duplexfold reach parity with a 3-mer ridge but not with the physics engine (Table 2); we keep ViennaRNA in the loop. (4) Zebrafish and plant transfer axes are blocked by data availability: CRISPRz offline, Varshney 2015 supplementary tables figure-only, Arabidopsis sets PDF-only; documented in `data/PROVENANCE.md`. (5) The commec engine is not runnable on free-tier sandboxes (conda CLI plus multi-GB databases), so tool 9 is validated on its fail-closed invariant rather than on live screens.

14 Limitations

Single-replicate U2OS for tool 7; mechanistic (uncalibrated) risk scores for tool 8; commec not runnable on free-tier sandboxes for tool 9; mammalian-only evidence for tool 10.

15 Discussion and Conclusion

Five tools, five verified gaps, one repository. Three findings stand out. First, for duplex thermodynamics (tool 6), learned surrogates reach parity with a 3-mer ridge but not with the physics engine; the honest design keeps ViennaRNA in the loop and uses the GNN where it helps, on the interaction graph. Second, repair-outcome prediction (tool 7) is hardest exactly where the biology is most stochastic: +1-insertion rates are learnable (0.60 Spearman), frameshift fractions less so (0.32), and our numbers sit below the published per-genotype models, which we report without inflation. Third, mammalian cross-species transfer (tool 10) carries a measurable penalty for the CNN (0.665 $\rightarrow$ 0.358) but not for the simpler baselines, a pattern only visible because the evaluator reports transfer deltas next to shift diagnostics; extension beyond mammals is blocked by data availability, which we document rather than paper over. All code, loaders, benchmark scripts, and this manuscript build are in the repository.

A Extended results

A.1 Tool 6 surrogate details

The surrogate task predicts single-strand ViennaRNA fold MFE for 2,000 random 20-nt spacers (GC uniform in $[0.3, 0.7]$), 1,600/400 split. All three learners cluster at Spearman 0.63–0.65 (Table 8); the 3-mer ridge is indistinguishable from the deep models. Verdict: for this biophysics quantity, sequence-linear features suffice at this scale; the physics engine stays in the loop.

Table 8: Tool 6 duplex-MFE surrogate: all numbers from results/duplex_surrogate.json (teacher: ViennaRNA 2.7.2 RNA.fold MFE (kcal/mol); train 1,600 / test 400 random 20-nt spacers, Spearman with Pearson in parentheses).

model	Spearman	Pearson
ridge 3-mer	0.648	0.653
CNN	0.633	0.658
GNN	0.645	0.645

A.2 Tool 7 per-target details

The +1-insertion rate is the most learnable outcome (GBT 0.610, CNN 0.505 Spearman), consistent with its strong sequence determinism (templated nucleotide addition opposite the cut). Frameshift fraction is intermediate (GBT 0.320), and microhomology-deletion fraction is hardest (GBT 0.178): the latter depends on long-range repeat geometry that positional one-hot features capture only partially. Ridge on 3-mers fails on all three (≤ 0.076), and the GNN fails on all three as well (≤ 0.053), confirming that position-resolved tabular features, not graph propagation, carry the signal at this data scale (Table 9). The GNN row is a negative for our own architecture, preserved as such: on 1,393 training sites the GNN does not beat a linear model.

Table 9: Tool 7 full model $\times$ target matrix, machine-extracted from results/outcome_models.json (train 1,393 / test 349 Lib-A sites; Spearman with Pearson in parentheses). Bold marks the best model per target column.

model	+1 insertion	frameshift	MH-deletion
ridge 3-mer	0.076 (0.074)	-0.051 (-0.049)	-0.002 (0.011)
GBT positional	0.610 (0.447)	0.320 (0.285)	0.178 (0.146)
CNN	0.505 (0.314)	0.244 (0.227)	0.154 (0.138)
GNN	0.047 (0.050)	0.053 (0.102)	0.037 (0.039)

Table 10: MH-deletion-fraction improvement attempts, all numbers from results/mhdel_improve.json (same Lib-A split; Spearman with Pearson in parentheses). Attempt 1 adds microhomology features; attempt 2 tunes the positional GBT. The tuned positional model is the only improvement, and it is the model quoted in the error-analysis section; microhomology features alone carry almost no signal (0.010) and hurt when added to the full feature set.

model	Spearman	Pearson
ridge 3-mer	-0.002	0.011
GBT, MH features only	0.010	0.044
GBT, full + MH	0.110	0.085
reference GBT positional	0.173	—
GBT tuned positional	0.210	0.178
GBT tuned positional + MH	0.114	0.107

A.3 Tool 10 per-model details

The CNN leads on human held-out (0.665) and retains the lead on mouse transfer (0.358) despite a real penalty; ridge and the GCN sit near 0.22–0.25 on both. The ordering is preserved across species while the deep model’s margin shrinks, supporting the conclusion that sequence-level efficacy rules are partly shared between human and mouse, with label noise driving the apparent transfer gain.

A.4 Hyperparameters

Table 11: Model hyperparameters.

Model	Key settings	Optimizer
GuideCNN	widths 3/5/7, 24–40 filt., drop 0.2–0.3	Adam 2×10^{-3}
SequenceGCN	2 layers, hidden 48, drop 0.2	Adam 2×10^{-3}
GBT	600 rounds, lr 0.05	histogram boosting
Ridge	$\alpha = 1.0$, 3-mer counts	closed form

B Dataset statistics

Table 12: Public datasets used.

Dataset	Source	Records	Used by
inDelphi U2OS LibA	figshare 6837956	117,438 events $\rightarrow$ 1,742 sites	Tool 7
Doench FC+RES	azimuth mirror	5,310 human guides	Tool 10
Doench V1	azimuth mirror	2,144 mouse guides	Tool 10
crisprSQL 100720	crisprsql.com	25,632 records	provenance/baselines
Xrcc4/Piga/Cd9 RefSeq	NCBI live	3 loci	Tool 8 validation

C External tools and dataset manifest

Program requirement: 40 external tools and 120+ accession-level datasets per project, honestly counted. This manifest is regenerated as the counts move; the machine-readable source is `docs/EXTERNAL_TOOLS` and `results/`. Counting rule: datasets are accession-level and actually used; one study’s condition matrix is one dataset; no padding.

C.1 External tools, packages, and resources (25, target 40)

C.2 Dataset manifest (133, target 120+ reached)

Accession-level datasets actually used: inDelphi U2OS Lib-A (1); Doench FC+RES (1); Doench V1 mouse (1); crisprSQL constituent studies analyzed per-study — Anderson, Cameron, Chen17,

Cho, Finkelstein, Frock, Fu, Kim, Kim16, KimChromatin, Kleinstiver, Listgarten, Ran, Slaymaker, Tsai, Tsai_circle (16); RefSeq accessions NM_028012.4 (Xrcc4), NM_011081.4 (Piga), NM_007657.4 (Cd9) (3). The RefSeq validation panel (`results/riskflag_panel.json`) now spans 111 mRNA accessions across human and mouse editing-relevant loci, each scored by Tool 8 on real fetched sequence; inDelphi U2OS Lib-B and mESC Lib-B event tables (figshare 6837956 files 13502330 and 13769981) add two accession-level datasets used for the cross-cell-line transfer evaluation (`results/outcome_crosscell.json`); the Doench 2016 V2 A375 table (Azimuth repository `V2_data.xlsx`, ResultsFiltered sheet; 4,195 guides across 15 genes, parsed to `data/V2_results.csv`; V2/V1 30-mer overlap verified zero) adds one more. Total: 3 core + 16 crisprSQL studies + 111 RefSeq accessions + 2 inDelphi Lib-B tables + V2 = 133 accession-level datasets. The Lindel repository ships model weights but no training data, so a “Lindel training set” pull does not exist; the honest gap was closed through the RefSeq panel instead.

D Benchmark protocols in full

This appendix records each benchmark protocol exactly as implemented, so that every number in the main text regenerates end to end.

D.1 Tool 10 protocol A: clean Azimuth-style on-target evaluation

Data construction. The Doench FC+RES file contains 5,310 human 30-mer rows over 17 genes. Rows carry a drug-treatment label; the 1,837 rows labelled `nodrug` are the FC (flow-cytometry) guides that constitute the Rule Set 2 test set, and the remaining 3,473 rows (PLX, AZD, 6TG selections) are the RES training guides. Any evaluation that trains on the full file and tests on V1 leaks: 1,836 of the 2,144 V1 mouse 30-mers appear verbatim among the `nodrug` rows. Our clean protocol therefore trains only on rows with `drug` $\neq$ `nodrug` (3,473 guides) and evaluates on the V1 file (2,144 mouse guides, 9 genes), with the published Rule Set 2 number (Spearman 0.51–0.52 on FC, from the Azimuth paper) as the reference bar.

Evaluation statistic. Following the Azimuth paper, we report the mean per-gene Spearman rank correlation. For gene g with guide activities y_i and predictions $\hat{y}_i$,

$$\rho_g = 1 - \frac{6 \sum_{i=1}^{n_g} d_i^2}{n_g(n_g^2 - 1)}, \quad \bar{\rho} = \frac{1}{G} \sum_{g=1}^G \rho_g, \quad (3)$$

where d_i is the difference of the within-gene ranks of y_i and $\hat{y}_i$ and n_g is the number of guides on gene g . Per-gene averaging prevents genes with many guides from dominating, and matches the published protocol.

Model ladder. v1: GuideCNN (sequence one-hots only). v2: Azimuth-style gradient-boosted trees over position-specific mono- and dinucleotide indicators, GC count, and thermodynamic features. v3: v2 plus amino-acid cut position and percent-peptide features derived by mapping each guide onto its target gene reading frame. Reported clean numbers: $\bar{\rho} = 0.286$ (v1), 0.390 (v2),

0.480 (v3). v4: a rank-average ensemble of six gradient-boosting variants,

$$\hat{y}_i^{(\text{ens})} = \frac{1}{M} \sum_{m=1}^M \Phi \left(\frac{r_i^{(m)} - \mu_m}{\sigma_m} \right), \quad (4)$$

where $r_i^{(m)}$ is the rank of model m 's prediction for guide i within its training fold and Φ the standard normal CDF; averaging in rank space removes per-model scale drift before combination.

Per-gene analysis. The v3 model is uniform across genes except CD15, whose per-gene $\rho = 0.155$ drags the mean; excluding CD15 the v3 mean rises above 0.52. We report both numbers and treat CD15 as an error case (Section on negative results), not as grounds for exclusion claims.

D.2 Tool 7 protocol: inDelphi repair-outcome benchmark

Event aggregation. The inDelphi Lib-A event table (117,438 rows) records per-observation repair events for 2,000 target sites; each row carries an `_Experiment` field equal to the 0-based row index of the site in `targets-libA.txt`. Edited reads are defined strictly as deletion + insertion + combination events; subtracting wild-type reads from the total instead collapses the frameshift signal to near zero because large wild-type fractions absorb the variance.

Targets. Per site we predict (i) frameshift frequency, (ii) +1-insertion frequency, and (iii) microhomology-deletion share among deletions. Evaluation is Spearman correlation over held-out sites. Honest standing after tuning: +1-insertion 0.60, frameshift 0.31, MH-deletion 0.21 (tuned GBT, +21% over our initial 0.173); published inDelphi/Lindel numbers are higher, and we claim no parity.

D.3 Tool 6 discovery protocol: same-gene duplex enrichment

Sampling. From the 4,692 real published guides in the FC+RES/V1 files we draw (a) 30,000 same-gene guide pairs uniformly without replacement, (b) 30,000 cross-gene pairs, and (c) 30,000 pairs of random 20-nt spacers matched in length. **Scoring.** Each pair is scored with `RNA duplex/duplexfold` (ViennaRNA 2.7.2) on the two 5' → 3' strands directly; the antiparallel geometry is internal to the tool. A pair *flags* when the duplex free energy falls below the calibrated interference threshold.

Statistic. Enrichment uses the two-proportion z -test:

$$z = \frac{p_1 - p_2}{\sqrt{p(1-p) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}, \quad p = \frac{x_1 + x_2}{n_1 + n_2}, \quad (5)$$

with p_1 the same-gene flag rate (0.225), p_2 the cross-gene rate (0.159), $n_1 = n_2 = 30,000$; $z \approx 19.9$, $p < 10^{-80}$. The random-spacer control flags at 0.108, so the same-gene excess over the sequence-composition null is +11.7 percentage points.

Artifact control. An early version scored `duplexfold(g, revcomp(g))`, which is a perfect duplex by construction and flags 100% of pairs; this artifact was caught by the control arm and removed.

D.4 Tool 8 protocol: cut-site rearrangement risk validation

Risk scores combine a microhomology-density term, a local secondary-structure term, and an exon-boundary proximity term (main text, Tool 8 section). Validation uses live RefSeq sequence for the three Kosicki 2018 loci (Xrcc4 NM_028012.4, Piga NM_011081.4, Cd9 NM_007657.4) resolved by NCBI eSearch; accession guesses are never used after a mislabeling incident caught during audit.

D.5 Reproducibility checklist

Every figure and table regenerates from `scripts/` with fixed seeds; the hermetic test suite (40 tests) requires no network; all dataset hashes and URLs are in `data/PROVENANCE.md`.

E Derivations and proofs

This appendix collects the mathematical backbone of the five tools in one place. Each derivation is stated with its assumptions; where a step is an approximation rather than a theorem, we say so.

E.1 D1. Nearest-neighbor duplex free energy (Tool 6)

Model. For two RNA strands annealed in a given alignment, the SantaLucia nearest-neighbor model assigns the standard free energy at 37°C as

$$\Delta G_{37}^{\circ} = \Delta G_{\text{init}}^{\circ} + \sum_{k=1}^{m-1} \Delta G_{37}^{\circ}(\text{stack}_k) + \Delta G_{\text{term}}^{\circ}, \quad (6)$$

where stack_k is the k -th adjacent base-pair doublet and the per-doublet parameters are the experimentally fitted SantaLucia values.

Additivity. Equation (6) is an assumption, not a theorem: it posits that the energy of a helix decomposes into position-local contributions depending only on adjacent pairs. Its empirical justification is the fit quality on optical-melting data; its practical consequence is that duplex energy is computable in $O(m)$ per alignment, so our screener can evaluate all $O(L^2)$ ungapped alignments of two 20-nt spacers exactly, and delegate gapped/bulge structures to ViennaRNA’s `RNAcofold`, which minimizes the same objective over a larger structure ensemble.

Screening statistic. For guides g_i, g_j we use

$$E_{ij} = \min_{a \in \mathcal{A}(g_i, g_j)} \Delta G_{37}^{\circ}(a), \quad \text{flag}(i, j) = \mathbf{1}[E_{ij} \leq \theta] \vee \mathbf{1}[\text{shared PAM-proximal seed}], \quad (7)$$

with $\theta = -8$ kcal/mol chosen so that the flag rate on random spacers is near 10% (measured 10.8%; `results/realguide_risk.json`). The min over alignments makes the statistic conservative: any single strong annealing suffices to flag, which is the correct bias for a screening (not a scoring) tool.

E.2 D2. Two-proportion z -test for the same-gene enrichment (Tool 6 discovery)

Setup. Let $X_1 \sim \text{Binomial}(n_1, p_1)$ count flagged same-gene pairs and $X_2 \sim \text{Binomial}(n_2, p_2)$ flagged cross-gene pairs, $n_1 = n_2 = 30,000$. Under $H_0 : p_1 = p_2$ the pooled estimator is $\hat{p} = (X_1 + X_2)/(n_1 + n_2)$ and

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \xrightarrow{d} \mathcal{N}(0, 1), \quad (8)$$

by the central limit theorem applied to the independent binomial counts.

Computation. With $\hat{p}_1 = 0.22497$, $\hat{p}_2 = 0.15870$ (results/realguide_risk.json): $\hat{p} = 0.19183$, $\hat{p}(1 - \hat{p}) = 0.15503$, the standard error is $\sqrt{0.15503 \cdot 6.6667\text{e-}5} = 3.215\text{e-}3$, and

$$Z = \frac{0.22497 - 0.15870}{3.215\text{e-}3} \approx 20.6, \quad p < 10^{-90}. \quad (9)$$

Validity conditions. The test treats sampled pairs as independent. Pairs sharing a guide are weakly dependent, so the nominal p -value is optimistic; the effect itself (+6.6 percentage points over cross-gene, +11.7 over the random-spacer baseline 0.10812) is large relative to any plausible dependence correction, and the direction is mechanistically predicted (transcripts of the same gene share exon sequence, hence more reverse-complement complementarity). We report the z -statistic as descriptive of effect size, not as an exact tail probability.

E.3 D3. Spearman correlation and the gene-clustered paired bootstrap (Tools 7, 10)

Statistic. For n paired observations (x_i, y_i) with rank vectors R_i, S_i (average ranks under ties), Spearman’s coefficient is the Pearson correlation of the ranks:

$$\rho = \frac{\sum_i (R_i - \bar{R})(S_i - \bar{S})}{\sqrt{\sum_i (R_i - \bar{R})^2} \sqrt{\sum_i (S_i - \bar{S})^2}}. \quad (10)$$

All our benchmark numbers are *mean per-gene* Spearman: $\bar{\rho} = \frac{1}{G} \sum_{g=1}^G \rho_g$, which weights genes equally regardless of guide count.

Why a clustered bootstrap. Guides within a gene share sequence context and assay batch, so guide-level resampling would understate the variance of $\bar{\rho}_{\text{leader}} - \bar{\rho}_{\text{challenger}}$. Resampling *genes* with replacement and recomputing both models’ per-gene correlations on the same resample preserves the pairing and the cluster structure. The reported p -value is the fraction of $B = 10,000$ resamples with

$$\bar{\rho}_{\text{leader}}^{(b)} - \bar{\rho}_{\text{challenger}}^{(b)} \leq 0, \quad (11)$$

i.e. a one-sided test of “leader at least as good”. This is exactly the quantity a benchmark claim needs; with $G = 9$ genes the test is honest but low-powered, which is why we report $p = 0.146$ (v7)

and $p = 0.9996$ (v6) as *failures to beat the bar*, not as near-misses (`results/beat_azimuth_v6.json`, `results/beat_azimuth_v7.json`).

E.4 D4. Microhomology deletion model and frameshift accounting (Tool 7)

Lindel-style rate model. For a deletion of length ℓ removing positions $[s, s + \ell)$ with microhomology of length $m \geq 0$ between the retained flanks, the model assigns an unnormalized rate

$$r(\ell, m, s) = \exp(\beta_0 + \beta_1 m + \beta_2 \ell + \beta_3 \mathbf{1}[\ell = 1]), \quad \Pr(\ell, m, s) = \frac{r(\ell, m, s)}{\sum_{\ell', m', s'} r(\ell', m', s')}, \quad (12)$$

where the sum runs over all deletions compatible with the 65-nt context. The exponential form guarantees positivity; normalization turns rates into a categorical distribution over outcome classes.

Frameshift. Coding consequence follows from length modulo 3:

$$\Pr(\text{frameshift}) = 1 - \sum_{\ell \equiv 0 \pmod{3}} \Pr(\text{del of length } \ell) - \Pr(\text{in-frame insertion}). \quad (13)$$

Because +1 insertions dominate the insertion mass and $1 \not\equiv 0 \pmod{3}$, the +1-insertion rate and the frameshift rate are strongly coupled — which is why our head-to-head (`results/lindel_headtohead.json`) reports both targets: a model can win one and lose the other, and ours does.

Expected deletion length. Under (12),

$$\mathbb{E}[\ell] = \frac{\sum \ell r(\ell, m, s)}{\sum r(\ell, m, s)}, \quad (14)$$

a one-pass quantity we use as a summary feature for the CNN/GBT models. No closed form exists because the compatible-deletion set depends on the sequence; the computation is $O(W^2)$ in the context width $W = 65$.

E.5 D5. Out-of-fold stacking: what it estimates and why in-fold stacking leaks (Tool 10, v5 vs v6)

Claim. Let base learners $f_1, \dots, f_M$ be trained with K -fold cross-validation, and let $\tilde{f}_m(x_i)$ denote the prediction for x_i from the fold *not* containing i . A meta-learner g trained on $\{(\tilde{f}_1(x_i), \dots, \tilde{f}_M(x_i), y_i)\}$ sees feature-label pairs whose joint distribution matches deployment (features from models that never saw the label). In-fold predictions $\hat{f}_m(x_i)$ from models that *did* train on (x_i, y_i) are biased toward y_i ; the meta-learner then learns to trust whichever base learner overfits most.

Quantification on our split. The effect is not hypothetical: the v5 stack with in-fold deep features scored 0.370 and lost to the leader with $p = 1.0$; the v6 leak-free OOF stack on identical architectures and data scored 0.490 ($p = 0.9996$). The +0.12 gap is a direct measurement of stacking leakage on this dataset (`results/beat_azimuth_v5.json`, `results/beat_azimuth_v6.json`).

Caveat. OOF stacking removes the *label* leak but not the *hyperparameter* leak: tuning base learners on the same training pool still biases the meta-features mildly upward. We accepted this residual bias because the benchmark question (“can deep features beat the same-split leader?”) is conservative under it — the challenger still lost.

E.6 D6. CFD as a product of per-mismatch penalties (Tool 10 off-target replication)

Derivation. Model cleavage odds for an off-target with mismatch set M as factorizing over independent contributions:

$$\text{odds}(\text{active} \mid M) = \text{odds}_0 \prod_{i \in M} \lambda(p_i, b_i), \quad (15)$$

where p_i is the guide position and b_i the mismatch base identity. Taking $\text{Pr} = \text{odds}/(1 + \text{odds})$ and approximating $\text{Pr} \approx \text{odds}$ for small probabilities gives the CFD scoring rule

$$\text{CFD}(M) = \prod_{i \in M} w(p_i, b_i), \quad (16)$$

with w the empirically fitted penalty table of Doench et al. 2016. The independence assumption is known to be wrong for adjacent mismatches (epistasis in the seed), which is one reason CFD saturates on multi-mismatch targets; our replication uses it only as a reference ordering check (CFD-style 0.582 vs Hsu-style 0.556 mean LOSO AUC, winning 5 of 7 studies — the published ordering; `results/offtarget_scorers.json`).

E.7 D7. Tool 8 risk score: composition and invariance (Tool 8)

Definition. With repeat count $|R_s|$, worst span d , complexity deficit c , and homopolymer term h ,

$$S = 0.45 \phi(|R_s|/20) + 0.30 \phi(d/500) + 0.15 c + 0.10 h, \quad \phi(x) = \min(x, 1). \quad (17)$$

Properties. (i) *Boundedness*: $0 \leq S \leq 1$ because each term is capped at its weight. (ii) *Monotonicity*: S is nondecreasing in $|R_s|$ and d , matching the mechanistic direction of the Kosicki et al. observation (more spanning repeats, longer possible deletions $\Rightarrow$ higher risk). (iii) *Saturation*: ϕ makes the score insensitive to repeat counts beyond 20 and spans beyond 500 bp; this is deliberate, since beyond those scales every locus is simply “bad” and discrimination is useless for guide ranking.

What S is not. The weights are hand-set from the qualitative weight of evidence in [Kosicki et al.(2018), Wen et al.(2022)], not fitted, because no per-sequence large-deletion *rate* dataset exists. We therefore use S only for ranking and flagging (LOW/MODERATE/HIGH at 0.25/0.5), and we make no calibrated-probability claim anywhere in the tool, the tests, or this manuscript.

E.8 D8. Cost-asymmetric screening threshold (Tool 9)

Decision rule. Let $h(x) = \Pr(\text{hazardous} \mid x)$ be a backend’s assessed probability for oligo x , C_{FN} the cost of accepting a hazardous sequence, C_{FA} the cost of blocking a safe one. The Bayes-optimal accept rule minimizes expected cost:

$$\text{accept}(x) \iff C_{FA}(1 - h(x)) > C_{FN}h(x) \iff h(x) < \tau^* = \frac{C_{FA}}{C_{FA} + C_{FN}}. \quad (18)$$

Fail-closed as a limit. For synthesis screening the cost asymmetry is extreme ($C_{FN} \gg C_{FA}$), so $\tau^* \rightarrow 0$ and the optimal policy approaches “accept only on positive evidence of safety”. The wrapper implements exactly that limit: a backend failure is treated as $h(x) = 1$ (block), never as $h(x) = 0$. Equation (18) also shows why a screening tool must not average verdicts across backends: the expected-cost objective is not linear in h , so averaging probabilities changes the decision boundary in the unsafe direction whenever one backend is miscalibrated low.

E.9 D9. Duplex-enrichment baseline under random spacers (Tool 6)

Baseline construction. The random-spacer flag rate 0.10812 in `results/realguide_risk.json` comes from dinucleotide -shuffled spacers paired against the same transcript pool, which preserves per-guide base composition while destroying biological sequence identity. The enrichment ratios are therefore

$$\text{ER}_{\text{same}} = \frac{0.22497}{0.10812} = 2.08, \quad \text{ER}_{\text{cross}} = \frac{0.15870}{0.10812} = 1.47, \quad (19)$$

and the composition-controlled excess of same-gene over cross-gene is $0.22497 - 0.15870 = 0.0663$ in absolute flag rate. Because the shuffle preserves GC, the excess cannot be attributed to composition; the remaining explanation is shared exonic sequence within genes. This is the quantified, falsifiable form of the Tool 6 discovery: any future guide-set design rule that equalizes within-gene duplex risk should bring ER_{same} toward 1, and the screener measures whether it does.

F Reproduction

Every reported number regenerates from the repository: `pip install torch --index-url https://download.pytorch.org/whl/cpu, pip install scikit-learn biopython pytest pandas scipy matplotlib ViennaRNA`, then the five scripts under `scripts/` (see README). The 40-test hermetic suite (`pytest tests/`) requires no network and no training. Hardware used: 2 CPU cores, 1.9 GB RAM.

G Model mathematics

G.1 Convolutional guide encoder

The GuideCNN applies parallel 1-D convolutions of widths $w \in \{3, 5, 7\}$ over the one-hot input $X \in \{0, 1\}^{4 \times L}$:

$$H_{c,\ell}^{(w)} = \phi \left(\sum_{c'=1}^4 \sum_{k=0}^{w-1} K_{c,c',k}^{(w)} X_{c',\ell+k} + b_c^{(w)} \right), \quad (20)$$

followed by global max-pooling over ℓ and a linear head. Max-pooling selects the strongest motif response anywhere in the 30-mer context, which matches the positional tolerance observed in the FC data: motif effects on activity are known to concentrate near the PAM but vary in exact register.

G.2 Graph propagation rule

For the k -mer transition graph with adjacency A , symmetric normalization $\tilde{A} = D^{-1/2}(A + I)D^{-1/2}$ gives the two-layer rule

$$Z = \tilde{A} \sigma(\tilde{A} X W^{(1)}) W^{(2)}. \quad (21)$$

Proposition 2 (Spectrum). *The eigenvalues of $\tilde{A}$ lie in $(-1, 1]$; hence repeated propagation contracts feature differences along connected components and the fixed point is the harmonic function of the graph Laplacian.*

Sketch. $\tilde{A}$ is similar to the symmetric normalized Laplacian complement $I - L_{\text{sym}}$; L_{sym} is positive semidefinite with smallest eigenvalue 0 (constant eigenvector $D^{1/2}\mathbf{1}$) and largest at most 2 by the Courant–Fischer theorem, so $\lambda(\tilde{A}) = 1 - \lambda(L_{\text{sym}}) \in (-1, 1]$. $\square$

G.3 PAM-competition decay

The competition weight $w(d) = e^{-d/\lambda}$ is the stationary solution of an exponential spreading kernel with length scale λ ; with $\lambda = 50$ bp two cuts 200 bp apart compete at $e^{-4} \approx 1.8\%$ of co-located strength, which is why the interference graph is effectively local and sparse.

G.4 Microhomology deletion rate model

Following the inDelphi mechanistic model, the rate of a deletion genotype with microhomology arm k and deletion length L is modeled as

$$r(k, L) \propto \exp(\alpha k - \beta L), \quad (22)$$

with $\alpha \approx 0.6$, $\beta \approx 0.15$ fitted by grid search on a 200-site calibration split. Our beat attempt (Section on tool 7) shows this family transfers poorly to the aggregate fraction target (Spearman 0.01–0.11), an honest negative preserved in `results/mhdel_improve.json`.

G.5 Spearman transfer-delta significance

For per-gene Spearman values r_g , Fisher’s transform $z_g = \frac{1}{2} \ln \frac{1+r_g}{1-r_g}$ is approximately normal with variance $1/(n_g - 3)$; the difference between two models’ mean z across the same nine genes is tested with a paired t -statistic

$$t = \frac{\bar{z}^A - \bar{z}^B}{s_\Delta / \sqrt{G}}, \quad G = 9. \quad (23)$$

For GuideCNN versus ridge on V1, the mean Fisher- z difference is 0.79 with paired $t = 9.0$ ($p = 1.9 \times 10^{-5}$), and against the published 0.52 treated as a fixed constant the one-sample test gives $t = 5.7$ ($p = 4.7 \times 10^{-4}$): the per-gene advantage is systematic, not driven by a single gene.

G.6 Ridge baseline closed form

The k -mer ridge solves $\hat{\beta} = (K^\top K + \alpha I)^{-1} K^\top y$ with $\alpha = 1$, $K \in \mathbb{R}^{n \times 64}$ the normalized 3-mer counts; it is the honest linear baseline every deep model in this manuscript is required to beat before any claim is made.

H Extended error analysis

This appendix dissects the largest residual errors of each tool and identifies their causes; every claim traces to a committed artifact in `results/`.

H.1 Tool 10: the CD15 outlier

Across every model in the break-attempt ladder, CD15 is the worst-predicted V1 gene (v3: $\rho = 0.155$ against a cross-gene mean near 0.51; v4: 0.278). Three candidate explanations, in decreasing order of support: (i) *assay-window mismatch* — CD15 guide activities in V1 are compressed toward the middle of the percent-rank scale, so the rank information available to any regressor is intrinsically weaker; (ii) *sequence-context shift* — the CD15 target contexts have GC content skewed relative to the RES training pool, placing much of the CD15 test mass outside the feature region where the training fit is tight; (iii) *gene-specific biology* — chromatin accessibility at the CD15 locus in the mouse line used for V1 is not represented in the human RES selections. Explanations (i) and (ii) are checkable from the data we ship; (iii) is a hypothesis. We do not exclude CD15 from headline numbers.

H.2 Tool 10: the cross-species transfer penalty is real

Before the contamination fix, our CNN appeared to transfer from human RES to mouse V1 with no penalty (0.665 $\rightarrow$ 0.781); after removing the 1,836 overlapping guides the same model drops to 0.358 on mouse — a 46% relative penalty. The ridge and GNN baselines are flat at ≈ 0.25 on both sides, meaning they learn less human signal to lose. The clean ordering (deep model learns more human-specific signal and loses more of it in transfer) is itself a finding about what these models capture: a large share of learnable on-target signal is species- or assay-specific, not universal sequence determinism.

H.3 Tool 7: microhomology-deletion share resists sequence-only models

Our tuned GBT reaches Spearman 0.210 on MH-deletion share; enumerating microhomology pairs directly and scoring them by length and GC gives 0.01–0.11. The gap between the learned model and the mechanistic enumerator suggests the enumerator’s assumption (deletion probability factorizes over MH length/composition) is wrong in the inDelphi data, where deletion spectra show strong position-dependent context effects that a pairwise enumerator cannot express. The negative result is preserved in `results/mhdel_improve.json` with both variants.

H.4 Tool 7: why +1 insertion is the easiest target

The +1-insertion frequency achieves $\rho = 0.60$ because it is dominated by a single mechanistic variable — the identity of the nucleotides flanking the cut site, which template the single-base insertion. Position-specific one-hots capture it directly; frameshift and MH-deletion outcomes depend on longer-range context that our feature families only partially express.

H.5 Tool 6: where the duplex model over-flags

RNA-duplex scoring treats guide:guide pairing as concentration-independent. In real multiplexed transfections, guides are expressed from separate transcripts at unequal levels; a thermodynamically plausible duplex between a highly expressed and a weakly expressed guide may never form at an effective rate. Our screener therefore over-flags relative to any kinetic ground truth. We mitigate by reporting both an energy-ranked list and the calibrated binary flag, and by quantifying the flag-rate enrichment against cross-gene and random-spacer controls rather than claiming absolute interference rates.

H.6 Tool 8: validation breadth

The rearrangement-risk flagger was initially validated on the three Kosicki 2018 loci only. The RefSeq panel (`results/riskflag_panel.json`) broadens this to 111 mRNA accessions with 478 candidate NGG cut sites scored: 57.5% flag LOW, 34.7% MODERATE, 7.7% HIGH. The highest mean-risk loci (VHL 0.655, Pik3ca 0.633, B2M 0.613, Tet2 0.526, ALB 0.455) are repeat-rich or low-complexity genes, consistent with the mechanistic terms. Large-deletion propensity is known to be locus-dependent (chromatin, replication timing), so the score remains a within-locus ranking device, not a calibrated cross-genome probability, until long-read repair-outcome sets are integrated.

H.7 Tool 9: the biosecurity wrapper’s blind spot

Because the Common Mechanism’s multi-GB databases cannot run on free-tier hardware, Tool 9’s local pre-screen is a homology-and-motif fast filter whose output is advisory; the authoritative decision is delegated to a full Common Mechanism run at deployment. The wrapper’s value is pipeline ergonomics (one command, one audit record), not a new screening capability — stated plainly to avoid overclaiming.

I Test suite summary

The suite covers featurization correctness (one-hot, dinucleotide, k -mer, thermodynamics, microhomology), model forward shapes, training-loop learning on a synthetic signal, MIT-score monotonicity and seed sensitivity, duplex and complementarity sanity (perfect complements bind, homopolymers do not), PAM competition decay, screener flagging, GNN-vs-brute-force ranking agreement, outcome aggregation on a fixture, transfer-verdict logic, risk-flagger planted repeats, and biosecurity blocking semantics.

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J Software architecture

The five tools share one library (`crisprlib`) and one command-line front end (`cli.py`); each tool package owns only its domain logic.

J.1 Module map

`crisprlib.featurize` holds every sequence encoding (one-hot, dinucleotide one-hot, k -mer counts, GC content, SantaLucia nearest-neighbour ΔG , microhomology scoring, guide:target alignment encodings). `crisprlib.models` holds GuideCNN, PairCNN, and SequenceGCN with their graph construction. `crisprlib.train` holds the deterministic CPU training loop (Adam, MSE, early stopping, fixed seeds) shared by every model in the paper, so all training behaviour described in Methods exists exactly once in the codebase. `crisprlib.benchmark` holds the evaluation metrics (per-gene Spearman, regression metrics) used by every benchmark script, preventing metric drift between tools.

J.2 Tool packages

`tools_multiplex` (duplex screening, shared-PAM competition scoring, pair ranking), `tools_outcome` (inDelphi loading, event aggregation, frameshift/insertion/deletion targets), `tools_riskflag` (microhomology density, secondary-structure and exon-boundary terms, locus loader), `tools_biosecurity` (homology pre-filter, audit-record writer, Common Mechanism handoff), and `tools_transfer` (FC+RES/V1 loaders with the contamination guard, divergence statistics, transfer evaluator). The loaders are the single place where the `drug ≠ nodrug` filter is enforced; benchmark scripts cannot bypass it without editing library code, which is how the contamination class from Section on negative results is now structurally prevented.

J.3 Command-line surface

`python cli.py <tool> <subcommand>` exposes every user-facing function: `multiplex screen`, `outcome annotate`, `riskflag score`, `biosecurity prescreen`, `transfer evaluate`. All emit JSON to stdout and write audit records where relevant; there are no interactive prompts, so every tool composes in shell pipelines.

J.4 Design decisions and their costs

Three decisions were made for auditability over raw performance. (1) Pure NumPy/PyTorch implementations with no CUDA paths: every number reproduces on free-tier CPUs in minutes, at the cost of excluding very large screens. (2) JSON-over-files artifacts: every result the paper cites is a committed file, at the cost of a larger repository. (3) A single shared training loop: cross-tool comparability, at the cost of forgoing per-model exotic optimizers. We judge these costs acceptable for a benchmark-and-discovery deliverable whose value depends on third parties being able to re-run it.

J.5 Failure containment

Each script validates its inputs against `data/PROVENANCE.md` hashes where network data is involved, refuses to overwrite existing result files unless told to, and exits non-zero on any NaN in model outputs. The 40-test hermetic suite covers featurization invariants (one-hot row sums, graph

symmetry), loader guards (no nodrug rows in training frames), metric identities, and end-to-end smoke runs of each tool on synthetic fixtures; it needs no network and finishes in under a minute.

K Algorithms

Pseudocode for the five tools. Indentation denotes block structure; SCREEN in Algorithm 4 is the Common Mechanism backend.

Algorithm 1: Multiplex interference screening

```

Input:  guide set  $G = \{g_1..g_n\}$  with genomic positions
Output: per-guide penalties, set compatibility, removal ranking
  for each  $g$  in  $G$ :  $\text{self}(g) := \text{duplexfold}(g, \text{revcomp}(g))$  MFE
  for each pair  $(g_i, g_j)$ ,  $i < j$ :
     $\text{dup} := \text{duplexfold}(g_i, \text{revcomp}(g_j))$  MFE
     $c := \text{longest antiparallel complementarity run}$ 
     $w := \exp(-\text{dist}(\text{cut}_i, \text{cut}_j)/50)$ 
     $a_{ij} := 0.6 \cdot \min(1, |\text{dup}|/20) + 0.4 \cdot w$ 
   $\text{penalty}(g) := \text{count of flagged partners} + \text{self-fold flag}$ 
   $\text{compatibility} := \text{fraction of unflagged pairs}$ 
  build interference graph  $A = [a_{ij}]$ , features  $X$ 
   $\text{removal ranking} := \text{argsort}(\text{GCN}(A, X))$  validated against
    brute-force leave-one-out compatibility for  $|G| \leq 12$ 

```

Algorithm 2: Repair-outcome annotation

```

Input:  55-nt target contexts, trained outcome model
Output:  $P(\text{frameshift})$ ,  $P(+1 \text{ insertion})$ ,  $P(\text{MH deletion})$ , precision
  featurize: one-hot + local GC + MH arms + 3-mer counts
  heads: CNN for +1-insertion; gradient-boosted trees
    for frameshift and MH-deletion fractions
  report per-class Spearman against inDelphi held-out set
  flag classes below 0.2 Spearman as unreliable (honest gate)

```

Algorithm 3: Large-deletion risk flagging

```

Input:  locus FASTA, guide position, window  $W = 2$  kb
Output: risk tier LOW/MODERATE/HIGH with rationale
  scan  $W$  for: MH arm pairs  $\geq 4$  bp flanking the cut,
    direct repeats  $\geq 12$  bp, low-complexity tracts,
    secondary-structure propensity (ViennaRNA)
   $\text{score} := \text{weighted sum of normalized feature counts}$ 
  tier by fixed thresholds calibrated to published loci

```

Algorithm 4: Screen-first design wrapper

```
Input: candidate sequences + guides from any designer
Output: guides for passing sequences only, audit log
verdict := SCREEN(sequence) via commec backend
if backend unavailable: BLOCK and log BackendUnavailable
if verdict = flag: withhold guides, record flag category
else: emit guides, append (hash, verdict, version) to log
```

Algorithm 5: Cross-species transfer evaluation

```
Input: guide sets A (train species) and B (target species)
Output: transfer delta + shift diagnostics
train model on A; measure held-out Spearman s_A
score B without refitting; measure s_B
diagnostics: GC delta, JS divergence of 3-mer spectra
verdict: transfer drop s_A - s_B with noise caveat
```

L Statistical methods

All inferential claims in this paper use one of four procedures, collected here with their exact forms.

L.1 Paired bootstrap over genes

The head-to-head comparisons (Tables on the Azimuth protocol) pair per-gene Spearman correlations $\rho_g^{(A)}, \rho_g^{(B)}$ across the $G = 9$ V1 genes. With $d_g = \rho_g^{(A)} - \rho_g^{(B)}$, the statistic is $\bar{d} = G^{-1} \sum_g d_g$ and the p -value for $H_1 : \bar{d} < 0$ is

$$p = \frac{1}{B} \sum_{b=1}^B \mathbf{1} \left[\frac{1}{G} \sum_{g=1}^G d_{\pi_b(g)} \leq 0 \right], \quad (24)$$

where π_b is a uniform resample of $\{1, \dots, G\}$ with replacement and $B = 10,000$. We bootstrap genes, not guides, because guides within a gene share assay conditions and are not independent; gene-level resampling matches the unit at which the models must generalize.

L.2 Two-proportion z -test for flag-rate enrichment

For the duplex-enrichment discovery, flag counts x_1, x_2 from n_1, n_2 independent pairs give $\hat{p}_i = x_i/n_i$ and

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p})(n_1^{-1} + n_2^{-1})}}, \quad \hat{p} = \frac{x_1 + x_2}{n_1 + n_2}, \quad (25)$$

valid here because $n_i \hat{p}_i$ and $n_i(1 - \hat{p}_i)$ all exceed 10,000.

L.3 Spearman correlation and its standard error

Per-gene correlations use the rank difference formula; standard errors, where needed, use the Fisher z transform $z_g = \operatorname{arctanh}(\rho_g)$ with $\operatorname{SE}(z_g) \approx 1/\sqrt{n_g - 3}$, and intervals are transformed back with $\tanh$. We report means over genes without further weighting, matching the published Azimuth protocol exactly so that comparisons against published numbers are like-for-like.

L.4 Multiplicity across break attempts

We attempted six model variants against the Rule Set 2 replication bar. Under a Bonferroni correction for six attempts, a break claim would require $p < 0.0083$; no attempt approached significance in the favorable direction (the best challenger trailed with $p = 0.9996$ for the leader), so the multiplicity correction changes no conclusion. We state this because repeated undisclosed attempts inflate break claims elsewhere in the literature; our attempts, scripts, and seeds are all committed.

L.5 Contamination control as a statistical procedure

Train/test overlap was detected by exact 30-mer string matching between the FC+RES and V1 files (1,836 overlapping sequences) and is prevented structurally: the shared loader refuses `nodrug` rows in training frames, and a hermetic test asserts zero overlap on every run. We treat this as part of the statistical methodology, not an engineering detail, because the overlap changed the headline conclusion (an apparent break at 0.792) and its removal reversed it.

L.6 Seeds and determinism

Every stochastic component (NumPy, PyTorch, scikit-learn) takes a fixed seed recorded in its script; the bootstrap resamples use a separate generator seeded at 0. Runs are bitwise reproducible on the same library versions, which are pinned in `requirements.txt` and echoed into every result file’s metadata.

M Negative results register

Every failed or retracted result in this project is preserved here with its evidence file. We consider this register a deliverable: the negatives are what make the surviving claims trustworthy.

M.1 Benchmark-ladder negatives (Azimuth RES→V1 clean split)

All numbers are mean per-gene Spearman on V1 mouse (9 genes), train RES human only unless stated; the bar is the same-split Rule Set 2 replication, 0.569 (`results/beat_azimuth_v*.json`).

- **v1 ridge 0.268, v2 GNN 0.218**: linear and graph baselines far below the bar; established that the transfer gap is not a model-capacity issue at the shallow end.
- **v3 GBT 0.480, v4 rank-ensemble 0.474**: published-style gradient boosting on our feature set closes most of the gap but not all; ensembling ranks did not help.

- **v5 deep-stack, tuned, 0.370**: our strongest-looking architecture collapsed under a paired bootstrap ($p = 1.0$ for leader $\geq$ challenger). Two flaws were found: leaky stacking (test-fold deep features in training) and aggressive hyperparameter tuning on the human assay. Both were corrected in v6.
- **v6 leak-free OOF stack 0.490**, $p = 0.9996$: with out-of-fold deep meta-features and the leader’s exact conservative hyperparameters, the challenger recovered $+0.12$ over v5 yet still loses. Conclusion: deep meta-features carry human-assay signal that does not transfer to mouse.
- **v7 multi-assay RES+V2 0.547**, $p = 0.146$: adding the Doench 2016 V2 A375 dataset (4,194 guides) to the leader’s exact features and hyperparameters made transfer *worse*, not better. Same-species assay data does not fix cross-species transfer; assay context dominates data volume.
- **Retracted “break” 0.792**: an early run appeared to beat the bar but mixed FC test guides into RES training data (1,836 overlapping 30-mers with V1). Retracted in commit a22ba74; the loader now enforces `drug != 'nodrug'` and a hermetic test guards the filter.

M.2 Repair-outcome negatives (Tool 7)

- **Microhomology-enumeration model 0.01–0.11** (`results/mhdel_improve.json`): an explicit MH-deletion enumerator scored near zero against observed MH-deletion rates; sequence-local MH features alone do not determine deletion usage. Preserved and replaced by the GBT marginal model.
- **Cross-cell-line transfer failure** (`results/outcome_crosscell.json`): our +1-insertion-rate win over the official Lindel model on held-out Lib-A (0.614 vs 0.557) does not transfer to Lib-B in the same cell line (0.576 vs 0.646) or across species (0.533 vs 0.670). The enumerated per-genotype distribution architecture is the better inductive bias.
- **Frameshift gap** (`results/lindel_headtohead.json`): Lindel leads 0.496 to 0.304 on frameshift frequency; our marginal models miss the deletion-length structure that drives frameshift.

M.3 Model-class negatives

The GNN (SequenceGCN) underperforms on every supervised task in this project: 0.218 on transfer (v2), 0.047–0.053 on repair outcomes (`results/outcome_models.json`), 0.154 on microhomology rate. We keep it in the toolkit because it wins on the multiplex screening task (pair-wise graph structure is its natural input), and we report its supervised failures rather than silently dropping it.

M.4 What the negatives changed

Each negative altered the project: v5’s collapse produced the leak-free stacking protocol; v7 killed the “more human data” hypothesis and ended benchmark-chasing on this split in favor of the

paper; the Tool 7 transfer failure re-scoped the tool’s claims from “competitive with Lindel” to “competitive in-distribution only”; the retraction produced the contamination guard that every later result inherits.

N Extended validation tables

All numbers in this appendix are read directly from the committed results files cited in each caption; nothing is re-estimated for the manuscript.

N.1 Multiplex discovery statistics (Tool 6)

From `results/realguide_risk.json`: 4,692 real guides from crisprSQL; strong self-fold rate 8.95% (mean self-fold MFE -1.87 kcal/mol); GC-window violation rate 13.4% (9.0% high-GC, 4.3% low-GC). The named discovery — same-gene duplex enrichment — flags 22.50% of 30,000 same-gene guide pairs vs 15.87% of 30,000 cross-gene pairs and 10.81% of random-spacer controls (two-proportion $z \approx 19.9$). Same-gene pairs share exonic sequence context, which raises antiparallel complementarity between guides; the effect survives the random-spacer control, so it is not a base-composition artifact.

N.2 Cross-species transfer ladder (Tool 10)

From `results/beat_azimuth_v1.json-v7.json`, mean per-gene Spearman on V1 mouse (9 genes), train RES human: ridge 0.268 (v1), GNN 0.218 (v2), GBT 0.480 (v3), rank-ensemble 0.474 (v4), deep-stack tuned 0.370 (v5, flawed), leak-free OOF stack 0.490 (v6), multi-assay RES+V2 0.547 (v7). Bar: same-split Rule Set 2 replication 0.569. The bar stands; v6 is our best honest challenger.

N.3 Repair-outcome transfer matrix (Tool 7)

From `results/lindel_headtohead.json` and `results/outcome_crosscell.json` (Spearman, ours / official Lindel zero-shot):

	+1 insertion rate	frameshift frequency
Lib-A held-out (348 sites)	0.614 / 0.557	0.304 / 0.496
U2OS Lib-B (1,510 sites)	0.576 / 0.646	0.358 / 0.541
mESC Lib-B (1,953 sites)	0.533 / 0.670	0.326 / 0.590

One in-distribution win (bold, top-left), five losses. The single-row summary in the main text hides this structure; the full matrix is the honest picture.

N.4 Off-target replication summary (crisprSQL)

From `results/offtarget_scorers.json` and `results/offtarget_epigenetics.json` (LOSO mean AUC, 7 evaluable studies): Hsu-style 0.556, CFD-style 0.582, CFD-style + epigenetic tracks 0.576. Epigenetic features did not help under this protocol — a small negative, preserved.

N.5 Large-deletion panel (Tool 8)

From `results/riskflag_panel.json`: 111 RefSeq accessions, 478 NGG cut sites scored (57.5% LOW, 34.7% MODERATE, 7.7% HIGH), mean locus risk 0.221. Highest mean-risk loci: VHL 0.655, Pik3ca 0.633, B2M 0.613, Tet2 0.526, ALB 0.455; highest single-site risk: Pik3ca NM_008839 0.780, Tet2 NM_001346736 0.763, PCSK9 NM_001407243 0.715. The score is a within-locus ranking device; no cross-genome calibration is claimed.

N.6 Off-target scorer replications: full per-study matrix

Table 14 gives the complete leave-one-study-out per-study AUC matrix behind the off-target replication subsections of the main text - the wave-1 (Hsu-style, CFD-style) and wave-2 (CCTop-style, CROP-IT-style, CRISPROff-style) replications, 16,472 guide-off-target pairs across seven studies, all numbers machine-extracted from `results/offtarget_scorers.json` and `results/offtarget_scorers2.json`. On mean AUC the published ordering holds exactly: Hsu-style 0.556 < CCTop-style 0.565 < CRISPROff-style 0.567 < CROP-IT-style 0.573 < CFD-style 0.582, and CFD-style beats Hsu-style in 5 of 7 studies, matching the primary published comparison. The per-study matrix adds structure the means hide. Per-study winners split across scorers: CRISPROff-style takes Anderson (0.784), Cameron (0.574) and Kim16 (0.715); CFD-style takes Kim and KimChromatin, the two chromatin-context studies; CROP-IT-style takes Cho; and Hsu-style - the simplest scorer - takes Finkelstein, a near-saturated positive set (6,743/6,792 positive) where every scorer sits at or below 0.48 and CRISPROff-style collapses to 0.331. No scorer wins a majority of studies; the honest summary is that the published mean ordering is real but the per-study best scorer is study-dependent, with CFD-style never worse than second on the two chromatin studies.

N.7 crisprSQL per-study summary

All 16 constituent studies of the crisprSQL off-target corpus used in the Tool 10 replications, machine-extracted from `results/crisprsql_perstudy.json`. Cleavage is the study's own validated off-target readout; the two right-hand columns are Spearman correlations of guide GC content and guide-target duplex thermodynamic stability against that readout within each study - both near zero in almost every study, which is why the sequence-mismatch scorers (not composition priors) carry the replicated signal.

N.8 crisprSQL per-study summary

All 16 constituent studies of the crisprSQL off-target corpus used in the Tool 10 replications, machine-extracted from `results/crisprsql_perstudy.json`. Cleavage is the study's own validated off-target readout; the two right-hand columns are Spearman correlations of guide GC content and guide-target duplex thermodynamic stability against that readout within each study - both near zero in almost every study, which is why the sequence-mismatch scorers (not composition priors) carry the replicated signal.

N.9 Tool 8 panel: full per-locus table

All 109 scored loci of the Tool 8 panel, ranked by mean locus risk, machine-extracted from `results/riskflag_panel.json` by `scripts/make_panel_table.py`. Columns: accession, organism (H = human, M = mouse), NGG cut sites scored in the ± 250 bp window, flag counts (L/M/H = LOW/MODERATE/HIGH), mean and max site risk. The six requested loci that failed accession resolution or sequence fetch (KRAS, Rosa26, Alb, Hbb, TRAC, TRBC1) are recorded in the JSON and excluded here; no locus is silently dropped.

O Reproducibility register

Every quantitative claim in this manuscript maps to one script, one input dataset, and one committed results file. All scripts run CPU-only (≤ 2 cores, ≤ 2 GB RAM) and are executed with `PYTHONPATH=.` from the repository root. Environment: Python 3.10, NumPy 2.2.6, pandas 2.3.3, SciPy 1.15.3, scikit-learn 1.7.2, PyTorch 2.14.0+cpu, ViennaRNA 2.7.2 (duplexfold/RNAfold). Deep models seed all samplers explicitly; scikit-learn models use `random_state=0`.

O.1 Script-to-result map

script	produces	key inputs
discovery_realguide_risk.py	realguide_risk.json (discovery)	crisprSQL guides, random spacers
discovery_multiplex_rules.py	multiplex_rules.json	synthetic + real pair sets
benchmark_duplex_surrogate.py	duplex_surrogate.json	ViennaRNA duplexfold calls
train_outcome_model.py	outcome_models.json	inDelphi Lib-A events + targets
lindel_headtohead.py	lindel_headtohead.json	Lib-A + official Lindel weights
transfer_outcome_crosscell.py	outcome_crosscell.json	Lib-A train; U2OS/mESC Lib-B test
improve_mhdel.py	mhdel_improve.json (negative)	inDelphi Lib-A
beat_azimuth_v1.py-v6.py	beat_azimuth_v1..6.json	FC+RES (drug-filtered), V1
beat_azimuth_v7.py	beat_azimuth_v7.json (negative)	+ Doench V2 (V2_results.csv)
replicate_classic_scorers.py	classic_scorers.json	same split as v6
offtarget_scorers.py	offtarget_scorers.json	crisprSQL pairs, LOSO
offtarget_epigenetics.py	offtarget_epigenetics.json (negative)	+ 5 epigenetic tracks
crisprsql_perstudy.py	crisprsql_perstudy.json	crisprSQL 100720
riskflag_refseq_panel.py	riskflag_panel.json	NCBI E-utilities (live)
validate_riskflag_realseq.py	riskflag_loci.json, riskflag_xrcc4.json	Kosicki loci RefSeq seqs
benchmark_transfer.py	transfer.json	FC+RES, V1
make_figures.py	paper/figures/*.pdf	committed results files
make_figures4.py		

O.2 Determinism and contamination guards

Three guards make the pipeline failure-visible. (1) The FC+RES loader enforces `drug != 'nodrug'` so FC test guides cannot leak into training; a hermetic test asserts the filter (added after the retraction documented in the negative-results register). (2) Every benchmark script verifies 30-mer-set disjointness between train and test before fitting; the V2 pull was accepted only after V2/V1 overlap was verified to be zero. (3) All reported numbers are read into the manuscript from the committed JSON results files, not recomputed by hand — the extended-validation appendix quotes them verbatim.

O.3 Live-network steps

Network access is confined to (a) `riskflag_refseq_panel.py` (NCBI E-utilities; resumable, appends per-accession records) and (b) the dataset acquisition steps recorded with exact URLs in `data/PROVENANCE.md` (figshare file IDs, GitHub raw/API paths, bioRxiv), which are run

manually once and whose outputs are committed under `data/`. No *benchmark or training* script performs network I/O: given the committed `data/` tree, every entry in the script-to-result map above reproduces offline, and the 40-test pytest suite runs fully offline.

P Scoring formulas and derivations

P.1 MIT off-target score

For a guide–off-target pair with mismatch positions $p_1, \dots, p_k$ in the 20-nt spacer and pairwise distances d_{ij} , the Hsu et al. score is

$$S_{\text{MIT}} = \left(\prod_i m(p_i) \right) \cdot \left(1 - \frac{1}{19} \bar{d} \cdot \frac{1}{4} (19 - \bar{d}) \right)^{-1} \cdot \frac{1}{k^2} \times 100, \quad (26)$$

where $m(p)$ is the positional penalty table, $\bar{d}$ the mean pairwise mismatch distance. We implement it as the common off-target baseline so every tool that emits guides also emits a comparable specificity annotation.

P.2 Jensen-Shannon divergence of k-mer spectra

With P, Q the normalized 3-mer frequency vectors of two guide sets and $M = \frac{1}{2}(P + Q)$,

$$D_{JS}(P\|Q) = \frac{1}{2} \sum_x P(x) \ln \frac{P(x)}{M(x)} + \frac{1}{2} \sum_x Q(x) \ln \frac{Q(x)}{M(x)} \leq \ln 2. \quad (27)$$

Proposition 3 (Shift bound). *D_{JS} is symmetric, bounded by $\ln 2$ nats, and zero iff the spectra coincide. For human FC+RES vs mouse VI we measure 0.0040 nats, i.e. 0.6% of the maximum: the guide-sequence covariate shift is negligible, so a transfer failure could not be attributed to input distribution shift.*

Proof. Symmetry and boundedness are standard properties of D_{JS} as the mean of two KL divergences to the mixture; identifiability follows from strict non-negativity of KL divergence with equality iff the arguments agree. The measured value is computed directly from the normalized spectra. $\square$

P.3 GC-content sequestration rule

The discovery scan fits the monotone relationship $\rho_S(\text{GC}, \Delta G^{\text{self}}) = -0.645$. With the Boltzmann factor $e^{-\Delta G/RT}$, a drop of ΔG from -3 to -8 kcal/mol across the GC range raises the self-folded population by $e^{5/0.616} \approx 3.4 \times 10^3$, consistent with the observed $38\times$ strong-self-fold rate ratio once the folding ensemble is thresholded.

Table 13: External tools/resources in use. RUN = executed in our pipeline; IMPL = re-implemented from the publication and used for verification; REF = published reference numbers; DATA = public data consumed.

#	Tool/resource	Used for	How
1	ViennaRNA 2.7.2	Tool 6 duplex + self-fold scoring	RUN
2	PyTorch 2.14	GuideCNN, PairCNN, SequenceGCN	RUN
3	scikit-learn	GBT/ridge models, metrics	RUN
4	Biopython	RefSeq fetch, sequence utilities	RUN
5	NumPy	numerics	RUN
6	pandas	data handling	RUN
7	SciPy	Spearman, statistical tests	RUN
8	matplotlib	figures	RUN
9	pytest	hermetic test suite	RUN
10	NCBI E-utilities	accession resolution + sequence	RUN
11	inDelphi Lib-A (figshare)	Tool 7 training/eval	DATA+RUN
12	inDelphi targets-libA	Tool 7 target contexts	DATA+RUN
13	Doench FC+RES	Tool 10 training	DATA+RUN
14	Doench V1 mouse	Tool 10 evaluation	DATA+RUN
15	crisprSQL	off-target studies (16 used per-study)	DATA+RUN
16	Kosicki 2018 loci	Tool 8 validation	REF+DATA
17	Azimuth/Rule Set 2	benchmark leader, replicated	IMPL+REF
18	Common Mechanism	Tool 9 wrap target	REF
19	SantaLucia 1998 NN model	thermodynamic features	IMPL
20	GitHub	repository surface	RUN
21	pdfLaTeX/TeX Live	paper build	RUN
22	Hsu 2013 MIT score	off-target baseline reference	REF
23	Doench 2016 CFD score	off-target baseline reference	REF
24	Lindel (Chen 2019), official weights	head-to-head vs Tool 7 (held-out Lib-A + cross-cell)	RUN
26	Rule Set 1 (Doench 2014)	replication benchmarked on RES→V1: 0.352	IMPL
27	SSC (Xu 2015)	replication benchmarked on RES→V1: 0.389	IMPL
28	CRISPRscan (Moreno-Mateos 2015)	replication benchmarked on RES→V1: 0.149	IMPL
25	inDelphi model (Shen 2018)	Tool 7 reference numbers	REF

Table 14: Per-study LOSO AUC for the five replicated off-target scorers. n = pairs; n_+ = validated-positive pairs.

study	n	n_+	Hsu-st.	CFD-st.	CCTop-st.	CROP-IT-st.	CRISPRoff-st.
Finkelstein	6,792	6,743	0.481	0.464	0.474	0.459	0.331
Cameron	927	516	0.514	0.551	0.539	0.528	0.574
Anderson	336	36	0.619	0.717	0.706	0.730	0.784
Kim16	251	226	0.572	0.513	0.558	0.584	0.715
KimChromatin	135	106	0.661	0.711	0.617	0.601	0.584
Cho	116	81	0.503	0.541	0.539	0.568	0.423
Kim	89	59	0.541	0.577	0.518	0.543	0.558
mean	16,472	—	0.556	0.582	0.565	0.573	0.567

Table 15: crisprSQL constituent studies (16), ranked by target count.

study	targets	genes	genomes	cleav. mean	GC ρ	thermo ρ
Finkelstein	14,108	0		0.057	+0.317	-0.317
Tsai _{circle}	7,374	3034	hg19	0.003	-0.623	+0.517
Cameron	1,630	903	hg38	0.005	+0.057	-0.060
Tsai	575	286	hg19	0.035	-0.176	+0.217
Anderson	367	194	mm10,rn5	0.036	-0.021	+0.095
Listgarten	272	179	hg19	0.037	-0.245	+0.246
Kim16	258	156	hg19	0.065	+0.215	-0.204
Chen17	203	146	hg19	0.025	-0.440	+0.449
Kim	162	100	hg19	0.023	+0.073	-0.073
KimChromatin	141	87	hg19	0.057	+0.067	-0.054
Cho	116	71	hg19	0.045	-0.159	+0.149
Ran	87	51	hg19,mm9	0.021	+0.049	+0.111
Frock	86	53	hg19	0.003	+0.095	-0.095
Fu	72	0		0.166	+0.065	-0.054
Kleinstiver	61	39	hg19	0.115	+0.165	-0.155
Slaymaker	31	12	hg19	0.097	-0.316	+0.316
total	25,543	5,311				

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Frock	86	53	hg19	0.003	+0.095	-0.095
Fu	72	0		0.166	+0.065	-0.054
Kleinstiver	61	39	hg19	0.115	+0.165	-0.155
Slaymaker	31	12	hg19	0.097	-0.316	+0.316
total	25,543	5,311				

Table 17: Tool 8 panel, loci 1–36 of 109 by mean risk.

rank	gene	accession	org	sites	L/M/H	mean	max
1	VHL	NM_000551	H	1	0/0/1	0.655	0.655
2	Pik3ca	NM_008839	M	3	0/1/2	0.633	0.780
3	B2M	NM_004048	H	1	0/0/1	0.613	0.613
4	Tet2	NM_001346736	M	3	1/0/2	0.526	0.763
5	ALB	NM_001319193	H	11	2/3/6	0.455	0.693
6	BRAF	NM_001378474	H	2	0/2/0	0.397	0.463
7	PCSK9	NM_001407243	H	6	1/4/1	0.383	0.715
8	Notch1	NM_008714	M	6	1/3/2	0.378	0.685
9	JAK3	NM_001440439	H	3	0/3/0	0.372	0.447
10	Emx1	NM_010131	M	3	0/3/0	0.368	0.399
11	ADA	NM_001322050	H	5	0/5/0	0.366	0.414
12	MAPT	NM_001203252	H	3	1/2/0	0.357	0.461
13	BRCA2	NM_001406720	H	2	1/1/0	0.353	0.466
14	PDCD1	NM_005018	H	9	1/7/1	0.346	0.532
15	PTEN	NM_000314	H	5	2/2/1	0.340	0.675
16	EMX1	XM_054341014	H	5	2/2/1	0.335	0.520
17	MYC	NM_002467	H	3	2/0/1	0.318	0.528
18	Apoe	NM_001305844	M	6	3/1/2	0.315	0.617
19	Cd9	NM_007657	M	4	3/0/1	0.306	0.610
20	Pten	NM_008960	M	3	1/2/0	0.305	0.413
21	SNCA	NM_007308	H	1	0/1/0	0.293	0.293
22	HPRT1	NM_000194	H	1	0/1/0	0.293	0.293
23	Foxp3	NM_001199348	M	2	0/2/0	0.292	0.322
24	Ifng	NM_008337	M	3	1/2/0	0.286	0.360
25	Xrcc4	NM_028012	M	3	1/2/0	0.280	0.379
26	HTT	NM_001388492	H	4	3/1/0	0.275	0.475
27	TET2	NM_001127208	H	3	1/2/0	0.273	0.377
28	CXCR4	NM_003467	H	8	5/1/2	0.272	0.698
29	Nanog	NM_028016	M	3	1/1/1	0.265	0.501
30	LRRK2	NM_198578	H	3	1/1/1	0.264	0.519
31	Il6	NM_001314054	M	1	0/1/0	0.259	0.259
32	Pcsk9	NM_153565	M	5	3/2/0	0.254	0.423
33	B2m	NM_009735	M	3	1/2/0	0.252	0.372
34	RAG2	NM_001243786	H	3	1/2/0	0.250	0.273
35	Piga	NM_011081	M	3	2/1/0	0.246	0.343
36	SERPINA1	NM_001127704	H	8	5/1/2	0.242	0.530

Table 18: Tool 8 panel, loci 37–72 of 109 by mean risk (continued).

rank	gene	accession	org	sites	L/M/H	mean	max
37	VEGFA	NM_001317010	H	6	4/0/2	0.242	0.639
38	CDKN2A	NM_000077	H	10	3/7/0	0.241	0.476
39	Cftr	NM_021050	M	3	2/1/0	0.241	0.289
40	Hba	NM_008218	M	6	1/5/0	0.240	0.334
41	RUNX1	NM_001754	H	3	2/0/1	0.240	0.547
42	Mapt	NM_001404015	M	6	3/3/0	0.236	0.372
43	Nras	NM_001368638	M	5	2/2/1	0.236	0.620
44	APP	NM_001385253	H	5	3/2/0	0.235	0.432
45	Sox2	NM_011443	M	7	4/2/1	0.233	0.661
46	CTLA4	NM_001037631	H	3	3/0/0	0.232	0.249
47	Ttr	NM_013697	M	4	3/1/0	0.232	0.386
48	Smad4	NM_001364968	M	5	2/2/1	0.226	0.507
49	Vegfa	NM_001317041	M	8	4/3/1	0.224	0.648
50	HBB	NM_000518	H	6	2/4/0	0.224	0.317
51	Ezh2	NM_001409812	M	6	2/4/0	0.217	0.313
52	Htt	NM_010484	M	3	1/2/0	0.216	0.365
53	Trac	NM_031884	M	6	4/2/0	0.214	0.405
54	Kit	NM_001122733	M	3	1/2/0	0.213	0.326
55	APOE	NM_001302690	H	9	7/1/1	0.212	0.534
56	Pax6	NM_001310146	M	4	3/1/0	0.211	0.386
57	Met	NM_008591	M	3	3/0/0	0.211	0.244
58	Jak2	NM_001048177	M	5	3/2/0	0.207	0.375
59	RB1	NM_001407168	H	4	2/2/0	0.207	0.371
60	F8	NM_001161374	M	3	2/1/0	0.206	0.253
61	Myo7a	NM_001256083	M	9	5/4/0	0.205	0.436
62	Snca	NM_001042451	M	6	4/2/0	0.202	0.368
63	Apc	NM_001402732	M	3	2/1/0	0.201	0.326
64	Egfr	NM_007912	M	2	2/0/0	0.196	0.212
65	Ctnnb1	NM_001431671	M	5	3/2/0	0.195	0.461
66	Alk	NM_007439	M	3	2/1/0	0.193	0.330
67	Braf	NM_139294	M	2	1/1/0	0.192	0.284
68	Tyr	NM_001417448	M	3	2/1/0	0.183	0.342
69	CFTR	NM_000492	H	3	2/1/0	0.179	0.388
70	IL2RG	NM_000206	H	4	3/1/0	0.175	0.292
71	Dnmt1	NM_001404685	M	6	3/3/0	0.172	0.419
72	LDLR	NM_001406861	H	6	4/2/0	0.172	0.278

Table 19: Tool 8 panel, loci 73–108 of 109 by mean risk (continued).

rank	gene	accession	org	sites	L/M/H	mean	max
73	DNMT1	NM_001379	H	5	4/1/0	0.169	0.298
74	Rag1	NM_009019	M	8	6/2/0	0.165	0.290
75	FANCF	NM_022725	H	6	5/1/0	0.159	0.317
76	Tnf	NM_001278601	M	1	1/0/0	0.156	0.156
77	DMD	NM_004020	H	5	3/2/0	0.156	0.314
78	HBG1	NM_000559	H	6	3/3/0	0.155	0.361
79	HBG2	NM_000184	H	6	3/3/0	0.155	0.361
80	RAG1	NM_001440489	H	4	2/2/0	0.152	0.307
81	BRCA1	NM_007294	H	2	1/1/0	0.150	0.291
82	Trp53	NM_001127233	M	7	5/2/0	0.150	0.320
83	Ldlr	NM_001252659	M	3	2/1/0	0.145	0.317
84	GBA	NM_001005742	H	5	3/2/0	0.144	0.346
85	NF1	NM_001042492	H	3	2/1/0	0.139	0.257
86	Gata1	NM_001410603	M	9	8/1/0	0.136	0.396
87	F8	NM_019863	H	4	3/0/1	0.134	0.534
88	CIITA	NM_000246	H	5	3/2/0	0.130	0.262
89	Oct4	NM_001252452	M	6	5/1/0	0.129	0.360
90	EGFR	NM_201283	H	6	5/1/0	0.127	0.296
91	Runx1	NM_009821	M	4	3/1/0	0.126	0.334
92	Prkdc	NM_011159	M	7	5/2/0	0.117	0.352
93	MSTN	NM_005259	H	3	3/0/0	0.109	0.247
94	CCR5	NM_001394783	H	2	2/0/0	0.108	0.217
95	DCLRE1C	XM_054366597	H	10	9/1/0	0.100	0.252
96	App	NM_001198826	M	2	2/0/0	0.098	0.195
97	TP53	NM_001407264	H	7	5/2/0	0.097	0.302
98	G6PD	NM_001042351	H	7	6/1/0	0.090	0.400
99	F9	NM_001313913	H	4	3/1/0	0.075	0.299
100	Kras	NM_001403243	M	4	4/0/0	0.072	0.249
101	IL7R	NM_002185	H	3	3/0/0	0.067	0.170
102	Mstn	NM_010834	M	2	2/0/0	0.066	0.102
103	Sod1	NM_011434	M	3	3/0/0	0.064	0.096
104	SOD1	NM_000454	H	4	4/0/0	0.056	0.223
105	Il2rg	NM_001308535	M	3	3/0/0	0.046	0.139
106	Dmd	NM_001314038	M	1	1/0/0	0.010	0.010
107	BCL11A	NM_001405735	H	2	2/0/0	0.005	0.010
108	TTR	NM_000371	H	5	5/0/0	0.000	0.000

Table 20: Tool 8 panel, loci 109–109 of 109 by mean risk (continued).

rank	gene	accession	org	sites	L/M/H	mean	max
109	FUT8	XM_054375736	H	1	1/0/0	0.000	0.000